UPPSALA
UNIVERSITET

Low-Resource CAPT: An Irish Application (WIP)

Peter Cady

Uppsala University
Department of Linguistics and Philology
Master Programme in Language Technology
Master's Thesis in Language Technology, 30 ECTS credits

May 24, 2026

Supervisors:
Johan Sjons, Uppsala University
Jim O' Regan, KTH Royal Institution of Technology

Abstract

[1] summarize the motivation
[2] problem overview
[3] experiment
[4] main findings and contributions.

Contents

Preface

Acknowledgements

I would like to thank my advisor Johan for his encouragement I would especially like to thank Jim O' Regan for all his thought-provoking insights, encouragement, and advice, which were crucial for me getting my bearings in a project which pushed me to grow well beyond my limits. The process was deeply meaningful to me. Thank you. I would like to thank my Irish professor Gregory Darwin for his commitment and engagement in promoting Irish literature and the Irish language. Thank examiner

Acronyms

AI	Artificial Intelligence 10
ASR	automatic speech recognition 3, 9, 10, 14–22, 26–28, 34–36, 38–40
BERT	Bidirectional Encoder Representations from Transformers 18
CALL	Computer-Assisted Language Learning 3, 10, 14, 15
CAPT	Computer-Assisted Pronunciation Training 3, 9, 10, 14–17, 38
CD	Correct Diagnostic 28
CER	character error rate 22, 23, 30
CNN	Convolutional Neural Network 18
CTC	Connectionist Temporal Classification 18, 35, 40
DE	Diagnostic Error 28
DER	Diagnostic Error Rate 28, 30
DNN	deep neural network 20
DPAD	diversified and preference-aware decoding 21, 34, 39
E2E	end-to-end 16–18, 20
FA	False Acceptance 27, 28
FAR	False Acceptance Rate 28, 30
FLEURS	Few-shot Learning Evaluation of Universal Representations of Speech 26, 27
FR	False Rejection 28
FRR	False Rejection Rate 28
G2P	grapheme-to-phoneme 22, 24, 26, 35
GCS	Google Cloud Service 27
GMM	Gaussian Mixture Model 17
GOP	goodness of pronunciation 16
HMM	Hidden Markov model 16, 17, 20
IPA	the International Phonetic Alphabet 13, 16, 22, 24–27
L1	first language 9, 12, 20, 21, 26, 39, 40
L2	second language 9, 12, 13, 17, 19–22, 26, 39, 40
LM	Language Model 39
LR	Logistic Regression 25, 27
MD	Mispronunciation Detection 9, 10, 16, 17, 20–22
MDD	Mispronunciation Detection and Diagnosis 3, 9, 10, 15, 19, 21, 22, 26–28, 38, 40
MFA	Montreal Forced Aligner 26, 35, 40
ML	Machine Learning 9, 18–20
MOOC	massive open online course 15
MT	Machine Translation 14, 21
NLG	Natural Language Generation 21
NLP	Natural Language Processing 3, 10, 11, 14
NN	neural network 11, 13, 21
OOV	out of vocabulary 26
P2P	phoneme-to-phoneme 21
ROVER	Recognizer Output Voting Error Reduction 19, 36
Seq2Seq	sequence-to-sequence 21, 26, 40
SSML	Speech Synthesis Markup Language 26

TA	True Acceptance 28
TL	Transfer Learning 14
TR	True Rejection 28
TTS	text-to-speech 9, 14, 19, 21, 22, 26, 34
WALS	the World Atlas of Linguistic Structure 10
WER	word error rate 20, 21
WFST	weighted finite-state transducer 40

1 Introduction

Language competency is crucial for securing work opportunities, accessing services, and exercising rights in society at large. As our surroundings become ever more digitally interconnected, these interactions are increasingly mediated by language technologies, for both good and ill. Such technologies are often identified as promising tools to promote language use and language learning, but are also implicated in the coalescence of online interaction around a few dominant languages such as English, complicating the glowing promises often offered by their proponents. Low-resource language communities open to language technologies are often left out of most recent technological advancements given the relative scarcity of resources they are faced with, necessitating approaches tailored to these limitations if the promise of cutting-edge language technologies is to be realized for these groups. The momentum of the pre-training/fine-tuning paradigm within automatic speech recognition (ASR) and Machine Learning (ML) in recent years offers a glimmer of hope to those working toward making accessible tools for such communities, enabling the use of more plentiful data to support tools for languages facing various levels of resource scarcity.

Promoting speech technologies for language learning could be of particular benefit for languages such as Irish which endeavor to propagate native models of speech effectively to motivated learners outside of traditionally Irish-speaking areas. Through Computer-Assisted Pronunciation Training (CAPT) applications built with careful use of existing data, we might find the scaleable, resource-conscious ways to support such communities and narrow the gap between tools built for digital behemoths such as English, and those built for languages without similar access to digital resources.

To this end we explore two potential avenues to overcoming the data limitations faced by Irish and other languages like it with respect to learner data: one, by using the data we *do* have by harnessing readily available monolingual data in a model ensemble; and the other, by imitating the data we *wish* we had with text-to-speech (TTS)-generated learner approximations to train a combined model with. These approaches are formulated as the following research questions:

1. To what extent does a resource-conscious ensemble of monolingual ASR models (Irish and English) offer improved discrimination of learner errors compared to a monolingual Irish model?

2. To what extent does a model trained on TTS-generated synthetic mispronunciation data offer improved discrimination of learner errors compared to a monolingual Irish model?

The unifying goal between these approaches is exploring the feasibility of overcoming the scarcity of phonetically annotated second language (L2) Irish learner data for Mispronunciation Detection (MD) and Mispronunciation Detection and Diagnosis (MDD) applications. One approaches the problem with ensembles of monolingual ASR models, leveraging more plentiful first language (L1) speaker data, and another using a schema of data generation by leveraging established TTS systems to approximate learner speech: a potential low-cost alternative to large-scale phonetic annotation. By exploring these approaches, we aim to illuminate possible ways forward for low-resource language communities interested in developing automated technologies for language learning and pronunciation training.

2 Background

In the following section we will outline the context motivating our current work within Natural Language Processing (NLP) and ASR for minority languages. We begin with the state of such languages in current research, highlighting some promising trends as well as some thusfar recalcitrant problems hindering more equal access to the rapid advances in language technology. We then proceed to outline the general research space of the current work: Computer-Assisted Language Learning (CALL) and CAPT before digging into the core task of CAPT systems: MD and MDD. We conclude with a technical overview of modern ASR systems used for such tasks, leaving the specific efforts of researchers to overcome data limitation inherent to low-resource languages in the next chapter, Related Work.

2.1 Irish & The Predicament of Minority Languages in Natural Language Processing (NLP)

Developing language technologies that can scale beyond the language they are designed for is no new goal for NLP research. The value of such a property is apparent: transferring an existing system seamlessly to another language could potentially save significant resources for language communities without the ability to fund such development themselves. Actually achieving this goal in practice, however, is no simple endeavor, typically requiring some level of linguistic awareness which is all-to-often lamentably absent (see Bender, 2011; Hedderich et al., 2021; Joshi et al., 2021, inter alia). For example, success in applying word-based n-gram approaches to a language rests on the language's level of inflectional morphology: the lower the better. Higher morphological complexity together with variations in word order raise data sparsity problems which n-gram approaches rooted in English struggle to handle (Bender, 2011). This should come as no surprise, given the relatively fixed word order and low levels of inflectional morphology present in English, but it illustrates a need for caution: systems developed for a given language may make implicit assumptions about language structure which do not generalize well. Linguistic typology can provide important clues as to what features are shared between the original development language and possible languages of extension for a system. Information of this kind has, for many of the worlds languages, already been gathered by linguists. Perhaps the most renowned database of typological information is the World Atlas of Linguistic Structure (WALS) (Dryer et al., 2024), a free, online resource currently boasting 152 chapters with detailed descriptions of 192 linguistic features spanning over 2,600 of the world's languages at the time of writing. By explicitly mobilizing linguistic knowledge already painstakingly gathered by linguistic typologists, we can identify where languages agree, where they differ, and hopefully identify some of the implicit, ungeneralizable assumptions that have hindered past efforts to create more accessible language technologies.

Despite claims of language-agnostic systems often touted by proponents of emerging Artificial Intelligence (AI) systems, these have often also fallen well short of such promise (Bender, 2011). The overwhelming majority of the world's languages still have no footprint in emerging language technologies (Joshi et al., 2021). In the past, building neural network (NN)-based language technologies demanded immense quantities of labeled data: a high bar of entry to language communities with limited if any access to such resources, and an ongoing issue which continues to stimulate a body of research dedicated to overcoming it (see Magueresse et al., 2020, for an overview). For languages where data availability is no obstacle, research and development can proceed unfettered by the prohibitive cost of curating datasets from scratch. As our daily lives grow increasingly integrated with the digital realm, language communities without the same support are obliged to switch to more digitally dominant languages (often English) to gain access to these new resources, narrowing the opportunities to engage with resources and services through the medium of their community language (Ní Chasaide et al., 2019). A particularly sobering taxonomy illustrating the states of languages facing such disparities is formulated by Joshi et al. (2021) and reproduced in Table 2.1, which outlines the challenges faced by languages in resource terms in the digital space, and how dominant a small group of languages are within it. Those at the bottom of the data availability scale at level 0, 'The Left-Behinds' have virtually no data of any kind available to support language technology development. Those in the middle at level 2 or 3 have some resources, net presence, and/or research communities supporting them, but critically lack in substantial quantities of the labeled data typically required for cutting-edge NLP tools. At the top, languages like English reap the lion's share of overall investment and development. Of particular note for the privileged few that find themselves at the top of the heap is their typological similarity, being drawn as they are from a few dominant language families (and even dominant branches within these larger families). This state of affairs constitutes a sort of typological echo-chamber for the cutting edge of NLP developments, a point which we will return to shortly. check table fix

Class Descriptions	**Example Languages**	**% of total languages**
0 The Left-Behinds: Virtually ignored in language technology. Exceptionally limited resources available, even with respect to unlabeled data.	Dahalo, Bora	88.17%
1 The Scraping-Bys: Some unlabeled data. With organized promotion and data collection, there is hope for improvement in coming years.	Fijian, Navajo	8.93%
2 The Hopefuls: Limited labeled data. Support communities help these languages survive, and there is promise for NLP tools in the near term.	Zulu, **Irish**	0.76%
3 The Rising Stars: Strong web presence and thriving community online. Lacking labeled data. Good potential for NLP tool development for these languages.	Indonesian, Hebrew	1.13%
4 The Underdogs: Much unlabeled data, and less but still significant labeled data. Dedicated investment from NLP communities.	Russian, Dutch	0.72%
5 The Winners: Dominant online presence with massive investment and resources.	English, German	0.28%

Table 2.1: Data availability & status taxonomy of languages adapted from work by Joshi et al. (2021).review paper rights. check rules for alterations and mention alterations (emphasizing irish, dropping some examples to save space etc.)

Despite some advantages not afforded other minority languages, Irish still finds itself struggling to maintain a footing in the the digital realm, placed by Joshi et al.

(2021) in class 2 of the taxonomy in Table 2.1. It enjoys ongoing investment by the Irish state, nominal status of Irish as the first national language of the Republic of Ireland, and research dedicated to its promotion (e.g. through the ABAIR initiative dedicated to developing speech technologies for Irish, see Chasaide et al., 2017). At the same time, it is a typological outlier in several respects: it is a verb-initial language with relatively complex inflectional morphology, and a distinct (though still Latin-based) orthography. Features like these put Irish at odds with many languages in the high-resource echo-chamber, complicating the ability to leverage cutting-edge models to linguistic features with no representation in a model's training data. Furthermore, though we have treated Irish as a single entity thus far, an important complication reveals itself in the discontiguous nature of the Irish-speaking areas, referred to as *the Gaeltacht* or *the Gaeltachtaí*(pl.). Each of the three main areas (i.e. Ulster, Connacht, and Munster 2.1) speak markedly distinct varieties of Irish, necessitating labeled data from each variant if these groups are to be adequately serviced by new technologies (Ní Chasaide et al., 2019). In the face of such limitations, today's data-hungry tools simply cannot be expected to perform to the same level on languages like Irish without the necessary resources. It should be noted that the pretraining/finetuning paradigm of recent massive multilingual models does mitigate this demand for data somewhat by leveraging unlabeled cross-lingual data, reducing the need for labeled data in the language being finetuned to (Hedderich et al., 2021; Joshi et al., 2021; Ranathunga et al., 2021), but for other languages without any labeled data to their name, this is a small comfort.

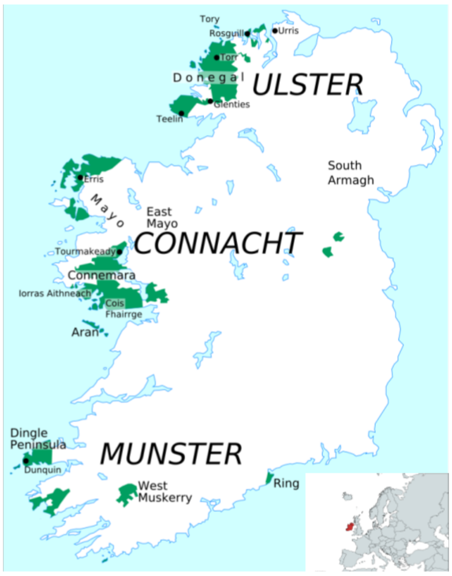

Figure 2.1: Area of the Gaeltachtaí colored in green. Originally uploaded by user Mahagaja and downloaded from Wikimedia Commons. Highlighted european map was created with mapchart.net

Though the hurdles facing the Irish language in the digital sphere are a relatively recent concern, the issues facing it in the real world are anything but. It is currently classified by UNESCO as being *definitely endangered* (Moseley and Nicolas, 2010), following centuries of varying rates of contraction due to encroachment by English. Irish survives as a community languages in the aforementioned Gaeltacht areas, though even there it is estimated that only 24% of inhabitants speak Irish on a daily basis (Ní Chasaide et al., 2015). Despite the state of Irish as a L1, it is comparatively strong as a L2 (Broin, 2014). A growing number of parents seek Irish-medium education for their children outside the Gaeltacht, and immersive summer courses remain popolar among adults looking to learn or reconnect with the language. This encouraging L2 engagement intersects with thorny issues of supply, however, as many of the teachers are not themselves native speakers with an accompanying native grasp of the structure and sound of the language (Ní Chasaide et al., 2015, 2019).

This limited native speaker model for L2 speakers is problematic for teaching pronunciation, particularly for the acquisition of some sound contrasts critical to disambiguating the Irish grammar. Perhaps chiefly among these, contrasts between the secondary articulation of consonants into *palatalised* and *velarised* variants play an instrumental role in a number of grammatical functions, such as in the formation of certain plurals and genitive marking (Broin, 2014; Gabriele, 2018; Snesareva, 2016; Stenson, 2020). Since the Roman alphabet does not provide symbols for this distinction, Irish orthography marks it via adjacent letters as seen in row (b) of table 2.2: so-called *slender* vowels ('i' and 'e') around a consonant denote *palatalisation*, while *broad* vowels ('a', 'o', and 'u') mark *velarisation*[1] (Stenson, 2020). Mutation effects are another pervasive element of the Irish grammar relying on sound alterations, the most common of which are *lenition* and *eclipsis*. Lenition, traditionally termed *séimhiú* (/ˈʃeːvʲuː/), is commonly marked with an 'h' following the lenited consonant as seen on Table 2.2, and originally denoted a weakening in the manner of articulation, though the relationship between consonants and their lenited versions is less immediately apparent now for some consonants (Stenson, 2020). Eclipsis, traditionally *úru* (/ˈʊɾˠuː/), involves replacing the original consonant with a nasalized or voiced version, and is denoted by appending the new sound character before the consonant being eclipsed2.2. These and other structural underpinnings of the language can have far-reaching implications for intelligability if not adequately mastered by students. Indeed, a study undertaken by Broin (2014) reveals that realisations of phonological details like those above have error rates of over 50% on average for urban speakers, with some as high as 82%—a stark departure from their gaeltacht counterparts which lie below 10%. Providing wider access to better native models of pronunciation—not to mention native models of morphology—could do much to close this gap, making mutual intelligability between gaeltacht and urban speakers more attainable.

Table 2.2: Examples use of a. consonant velarisation & b. palatalisation c. séimhiú & d. úru. Adapted from Ní Chasaide et al. (2015) with determining orthography in bold.check paper rights, elaborate on alterations (dropping one line, changing examples triggering séimhiú & d. úru slightly for some reason?)

	Orthographic	the International Phonetic Alphabet (IPA)	Translation
a.	bá**d**	/bˠaːdˠ/	'boat'
b.	bá**id**	/bˠaːdʲ/	'boats'
c.	do **bh**ád	/dˠɔ waːdˠ/	'your boat'
d.	ár **mb**ád	/ɛɾʲ mˠaːdˠ/	'our boat'

2.2 From Text to Speech: Approaches to Low-Resource Scenarios

With the rise of NN-based language technologies, the data-hungry nature of such tools underscores the urgency of addressing the kind of resource disparities outlined above. Making such tools accessible to languages without the same strong data foundations as English is an active area of ongoing research, though even within popular

[1] For initial consonants, the following vowel plays the determining role, for final consonants, the preceding vowel, and for medial consonants, the vowels flanking it.

languages such as English, non-standard domains and tasks types can constitute low-resource areas which lack suitable quantities of training data. These data disparities can be categorized along several dimensions, such as those proposed for NLP by Hedderich et al. (2021) as: availability of *task-specific labeled data* for the target language or domain, availability of *unlabeled* language- or domain-specific data, or the availability of *auxiliary* data. This latter kind of data is diverse, as it is data not directly labeled for the task at hand, but which can still be indirectly useful, from labels specific to another language/domain, to knowledge bases such as entity lists, or automated labels from Machine Translation (MT) systems (Hedderich et al., 2021).

To address the different dimensions of resource scarcity outlined above, various approaches have been developed which Hedderich et al. (2021) splits broadly into those which *generate additional labeled data*, and those employing *Transfer Learning (TL)*. Faced with limited gold-standard annotated data, researchers employ strategies of the former type to automatically produce labeled alternatives. These strategies can themselves be broadly grouped as *data augmentation*, where task-specific labeled data is used to make more labeled data, such as with Back-Translation for MT where a target-to-source translation model is used to obtain a synthetic parallel corpus from a monolingual target corpus (Ranathunga et al., 2021), and *distant supervision* which produces labels for existing unlabeled data, for example in Cross-Lingual Annotation Projection where a task-specific classifier is trained for a high-resource language, then projected onto text from a low-resource language using a paralell corpus. For TL, in contrast, instead of creating or extending task-specific training data, the focus lies on reducing the need for such data by leveraging models or learned representations from other languages/domains. This approach has been particularly successful in in recent years with the advent of models like BERT (Devlin et al., 2019) and Wav2Vec2 (Baevski, Zhou, et al., 2020) which are *pre-trained* on vast quantities of unlabeled data to then be *fine-tuned* for specific downstream tasks. This pretraining/fine-tuning paradigm can be particularly advantageous for languages or domains where labeled data is limited.

Although these strategies are commonly employed in NLP, analogous trends can be found in computer vision as well as speech to tackle similar limitations in data. TL and the aforementioned pretraining/fine-tuning paradigm has made strides with self-supervised models like Wav2vec2, outperforming previous state-of-the-art ASR models with 100 times less data, starkly reducing the demand for labeled speech (Baevski, Zhou, et al., 2020). Alongside TL, ensemble methods have also emerged as a promising tool for low-resource contexts. Here, specialist models with complementary attributes can be combined to perform better than any one of its constituents for novel tasks (Arunkumar et al., 2022; L. Deng and Platt, 2014; Fiscus, 1997; Gitman et al., 2023, inter alia). Various methods of data augmentation and data synthesis are also prevalent to improve ASR performance for low-resource languages, including voice transformations, where noise or other alterations are introduced to recordings to extend existing data, or generating synthetic audio data samples with a TTS system to bolster training data when authentic speech corpora are lacking (Bartelds et al., 2023; D. Zhang et al., 2022).

2.3 Computer-Assisted Language Learning (CALL) & Computer-Assisted Pronunciation Training (CAPT)

Despite efforts to the contrary, the rise of digital technologies is often implicated in the acceleration of already precpitous rates of decline for endangered languages (Ní Chasaide et al., 2015, 2019, inter alia). However, it is a trend that cuts both ways, as the same technologies that squeeze certain languages out of the digital realm are also making space for communities of language learners to come together towards their common goal through CALL platforms and massive open online course (MOOC)s. The increased presense of technology both in and outside the classroom brings with it broad implications for traditional pedgagogy, enabling more autonomous and flexible modes of learning for students (Chapelle, 2008) particularly for learners looking to autonomously improve their pronunciation via CAPT systems. This technological shift has increased the financial viability of courses for endagered or otherwise less commonly taught languages, allowing teachers to draw from a more geographically dispersed enrollment pool and provide courses otherwise impossible to offer. Despite this potential, tension between the technology and pedagogy underpinning CALL and CAPT systems remains a well-documented issue, and their increasing prevalence in the modern language learning landscape has renewed calls for greater collaboration between pedagogical and technical experts when designing these systems (Rogerson-Revell, 2021). This divide between technical experts and real-world application also crosscuts the aforementioned struggles of low-resource language communities in making the most of recent technological developments, though some languages such as Irish are still fortunate enough to have research groups working to bridge that gap (Ní Chiaráin, 2022; Ní Chiaráin et al., 2022).

The proliferation of CALL software for self-study such as Duolingo, Babbel, and Rosetta Stone proliferate, has renewed interest in the role of immediate and personalised feeback for pronunciation training in CAPT systems. More recent research indicates that language learners may need explicit and targeted feedback on their pronunciation in order to improve (Bajorek, 2017), despite the long-held understanding that error correction typically does not meaningfully influence acquisition (Krashen, 1984). Perhaps as a consequence of this established view, explicit pronunciation training has been notably absent from language classrooms in recent decades, and many CALL platforms carry on this legacy with binary right-or-wrong feedback mechanisms that do not make use of the potential for more effective, targeted feedback (Bajorek, 2017). The feedback is also frequently unexplained, making such binary judgements about pronunciation quality opaque to the student and thus more difficult to act upon. The ideal individualized, explaination enriched feedback would normally carry a steep price tag for the student of an individual tutor, say, but CAPT systems can potentially lower this barrier of entry considerably by automating the same kind of undivided feedback in a one-to-many form scalable to many geographically disperesed students at once.

Although the scope of this work does not evaluate the effectivness of any particular pedagogical application, we recognize the potential of CAPT systems and the need for pedagogical and linguistic awareness in their design to more fully deliver on their promise. To that end, we endeavor in this work to contribute with linguistically aware ASR strategies which can hopefully help bridge the gap between cutting-edge language technology and the needs of minority language communities, supporting pedagogically sound feedback to learners.

2.4 Pronunciation & Mispronunciation Detection and Diagnosis (MDD)

To realize the often untapped potential of CAPT and lay the groundwork for actionable feedback, the CAPT system must be able to identify deviations in student pronunciations from the target pronunciation and determine how it differs in ways interpretable to the student. To do this, we can start by specifying what we mean by pronunciation. In broad strokes, the human speech apparatus may be seen as a collection of subsystems which can emit signals over parallel channels (Engstrand, 2004, p. 173). Although speech is a continuous signal, we can map symbols of discrete speech sounds–phones–onto subsegments of this signal. These discrete units—the vowels and consonants that constitute words—are *segmental* features of speech. The prosody, intonation, stress, and other such elements of speech can be seen as superimposed on these segmental features, and are thus termed *suprasegmental* features of speech (Engstrand, 2004). Human cultures have an array of strategies for representing speech in written form, ranging from logographic writing systems like Chinese where one symbol represents one word, to different varieties of sound-based systems such as syllabic for Japanese hiragana or katakana, alphabetic like the Roman alphabet used here, or consonantal as in Semitic scripts (Jurafsky and Martin, 2025). Many, though by no means all, approaches to MD within CAPT focus on recognizing these segmental features in speaker recordings and comparing their phonetic transcriptions to a canonical pronunciation of speech for the target word (Korzekwa et al., 2022). For our purposes, we will similarly narrow our investigation of MD to first focus to detection of these segmental features, representing pronunciation as strings of phones using the IPA standard of phonetic notation. Though suprasegmental features can also play a crucial role in disambiguating meaning, investigating them is beyond the scope of this thesis.

Segmental recognition for MD can be broadly divided into two approaches: those which align speech signals to canonical segment sequences, and those which first extract segments—generally phonemes or phones—from speech, then aligning the extracted sequence of segments to a canonical sequence for comparison (Korzekwa et al., 2022). The process of aligning speech to text which characterizes the former approach is termed *Forced Alignment.* Classic pronunciation scoring generally focused on these Forced Alignment approaches, which typically relied on log-likelihood scores and log-posterior scores derived from Hidden Markov model (HMM)-based ASR model outputs (S. M. Witt, n.d.). The latter of these scores soon became the de facto standard due to its higher correlation with human assessments. Building further on this development, S. M. Witt (2000) introduced the widely used goodness of pronunciation (GOP) measure of pronunciation quality. The resultant score can be compared against a threshold value to determine how well the speaker's pronunciation matches the canonical model pronunciation (S. Witt and Young, 2014). The classic formulation of GOPs base form was given by S. M. Witt (2000) as the log of the posterior probability of any phoneme given an acoustic segment normalized by the length of this segment as given below by:

$$GOP_1(q_i) \equiv \log(P(q_i|O))/NF(O) \tag{2.1}$$

where O is the speech segment, and $NF(O)$ is the number of frames of segment O.

Although GOP and other posterior-based scores continue to be used for MD with modern ASR architectures (see Gong et al., 2022; Parikh et al., 2025, inter alia for examples), with the advent of conceptually simpler transformer-based end-to-end (E2E) models in the last decade, researchers have also pursued free-phone output sequence alignment to detect mispronunciation (Wu et al., 2021). This alternative circumvents the need for phone-level forced alignment of speech signals to reference transcriptions (Leung et al., 2019; Lo et al., 2020), and can be used to diagnose mispronunciation errors instead of simply detecting them by stopping at scoring pronunciation quality.

To detect mispronunciations at the segmental level, our approach starts with the former family of approaches outlined above: using forced alignment-derived phonemes as an intermediate step before comparing them to canonical strings. This focus on articulatorily meaningful units of speech could lay the ground for more informative and actionable feedback in any future CAPT extension, particularly in comparison to binary assessment strategies of simply identifying correct or incorrect pronunciation, but this focus introduces its own set of challenges. Recognizing phonemes requires fine-grained annotations at that level, which are difficult to obtain, particularly for L2 speech (D. Zhang et al., 2022). This framing of the MD problem is a data-scarce task to pursue, but the modern innovations in ASR architectures leveraging the aforementioned pre-training/fine-tuning paradigm could make it an increasingly feasible one, even for low-resource languages.

2.5 Speech and ASR

The critical technology underpinning segmental MD approaches is the ASR system transcribing spoken words into writing. ASR research spans many decades, having matured from its origins as a barely useful method of interface to the point where speech is the primary modality of interaction for popular forms of human-machine communication today (Yu and L. Deng, 2015). Some simpler tasks have been long since solved, such as simple yes-no recognition or digit recogniton (i.e. 0-9). Transcribing strings of phonemes or words like today's ASR models however, is far more complex, with many tens of thousands of possible vocabulary items to predict (Jurafsky and Martin, 2025). In recent decades, however, even these tasks have proven tractable under controlled conditions, as evidenced by the popularity of different commercial voice-controlled digital assistants such as Amazon's Alexa[2], Apple's Siri[3], or Google Assistant[4]. Modern ASR need to be robust to more than just vocabulary size, however. Jurafsky and Martin (2025) highlights other consequential dimensions of speech variability such as who the speaker is talking to; *read speech* such as that produced when humans talk to machines, is far easier to recognize than spontaneous *conversational speech* between two humans. The quality of the signal can also be a factor in the ease of recognition: speech recorded under optimal conditions in a recording studio will of course be easier to recognize than someone speaking at a distant microphone on a busy street. Finally—and perhaps most importantly for the current work—variation in speaker classes can impact robustness of an ASR when the speaker class lies outside what the system was trained on. Speaker class includes differences in *accent* or *dialect*, as is most relevant for our purposes, but also by age and sex/gender. Creating systems robust to all of this variability is by no means a solved problem, despite the notable progress in recent decades.

Prior to the turn of the century, HMM-Gaussian Mixture Model (GMM)-based models were the de facto standard for ASR, typically consisting of four main components: Signal processing and feature extraction, acoustic model, language model, and hypothesis search (Yu and L. Deng, 2015). These systems convert of an acoustic waveform into feature vectors by a signal processer, which are then combined by a decoder with a dictionary and language model or grammar network into a recognition network. Using this network, one can calculate the most likely word sequence given the waveform of the speech input (S. M. Witt, 2000). A number of toolkits have cropped up to support development of these systems, such as Kaldi (Povey et al., 2011), and later ESPnet (Watanabe et al., 2018) as E2E architectures became more prevalent. These

[2]https://alexa.amazon.com
[3]https://www.apple.com/siri/
[4]https://assistant.google.com/

toolkits provided libraries and useful recipes to create reproducible systems for an array of use cases, with Kaldi in particular becoming a very popular toolkit with which to develop ASR systems. Although these are still in use, the rise of multi-purpose ML libraries like Pytorch (Paszke et al., 2019) and Hugging Face (Wolf et al., 2020) have provided a simplified alternative to training modern ASR systems[5].

2.5.1 Wav2Vec 2.0

Following the rise of Transformers in ML generally, transformer-based ASR models such as *wav2vec 2.0* (commonly Wav2Vec2) have become increasingly popular for ASR tasks, including extracting phonemes from raw acoustic data. Similar to other E2E alternatives, it is conceptually simpler than leading models, while its self-supervised pre-training framework achieves competitive performance with a reduced dependance on labeled data. It consists of three main components: a Convolutional Neural Network (CNN)-based *feature encoder*, a Transformer-based *context network*, and a *quantization module*2.2. The feature encoder takes raw waveforms as input and outputs latent speech representations to be used both by the context network and the quantization module. The context network captures contextual information about the encoder output, while the quantization module divides the encoder output into discrete speech representations to be used as targets during self-supervised pre-training (Baevski, Auli, et al., 2020). During pre-training, a given proportion of feature encoder outputs (time steps) are masked, similar to the masked language model pre-training objective of Bidirectional Encoder Representations from Transformers (BERT) (Devlin et al., 2019). For each masked time step, the model then identifies the quantized speech representation produced by the aforementioned quantization module by solving a contrastive task.

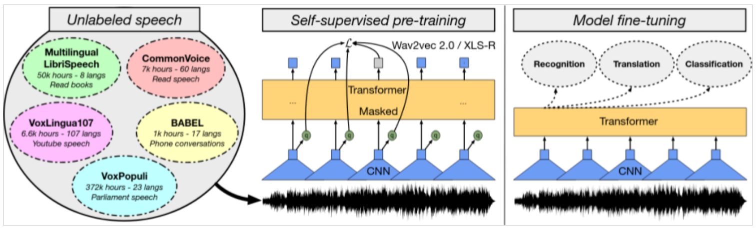

Figure 2.2: Illustration of the Wav2Vec2 framework adapted from Babu et al. (2021), no alterations

Using the above framework, Conneau et al. (2020) introduced the XLSR Wav2Vec2 model pretrained on multiple languages, with Babu et al. (2021) introducing several well known checkpoints of XLSR models pretrained on 53 languages at three different sizes of 300 million parameters, 1 billion parameters, and 2 billion parameters often short-handed as 300m, 1B, and 2B variants respectively. After pre-training on unlabeled data, these networks can be fine-tuned by adding a linear layer head of a size corresponding to the vocabulary of the task, and then optimized using Connectionist Temporal Classification (CTC) loss (Graves et al., n.d.) on a much smaller set of labeled data, achieving state-of-the-art results in scenarios where labeled data is scarce Baevski, Zhou, et al. (2020). These processes can be seen in Figure 2.2 adapted from Babu et al. (2021).

[5]It should be noted that ESPnet-EZ (Someki et al., 2024) provides a more modernized, python-focused interface similar to these generalist ML libraries while retaining some popular key features from Kaldi such as recipes, which have themselves become increasingly common in generalist libraries as well.

3 Related Work

In the following section, we will dive into some existing strategies for overcoming data scarcity in MDD and ASR more generally. These approaches can be broadly split into two groups: attempts to eliminated dependency on the scarce L2 data entirely, or through generating the desired L2 data through various synthesis or augmenting techniques(Korzekwa et al., 2021). This delineation echoes the approaches to data scarcity for low-resource languages outlined in section 2.2. Both avenues will be explored in the current work, specifically through:

1. ASR model ensembles, –and–
2. TTS-generated data augmentation

The former will serve as our exploration into knowledge transfer to circumvent the need for scarce L2 data, where the latter instead investigates the ability to overcome this scarcity directly through data synthesis.

3.1 ASR Ensembling Strategies

As covered more extensively in the previous chapter, the state of modern ML has seen a proliferation of models trained on massive amounts of data tuned to perform well on a variety of target tasks, but which can still struggle to perform satisfactorally outside their domain of expertise. One long-standing general strategy for overcoming the limitations of any single classifier model in ASR as in ML more generally is through *ensembling*. In general terms, an ensemble refers to a set of classifiers whose outputs are combined in some way (Dietterich, 2000). Doing so effectively requires that these classifiers be similar in function but complementary, and being diverse enough to cover each other's weaknesses (Hansen and Salamon, 1990). Originally, ensembles combined outputs of constituent models through Bayesian averaging, i.e. weighing the output of the model by the posterior probability of the output, giving each model what amounts to a weighted vote in the output (Dietterich, 2000). Despite this mathematically principled origin, it proved to be prohibitively time consuming to calculate for complex problems, and so alternatives were introduced which were more computationally tractable. A wide range of ensembling techniques have emerged over the years, including boosting, bagging, stacking, majority voting and confidence voting, all of which have proven robust methods of improving performance on a variety of tasks (see l. Deng et al., 2012; Fiscus, 1997; Gitman et al., 2023; Maclin and Opitz, 1997, inter alia). For ASR, these approaches have yielded improvements over singular models with combinations of different model architectures (Arunkumar et al., 2022; L. Deng and Platt, 2014), and different training data such as for specific languages or dialects (Agrawal et al., 2023; Gitman et al., 2023, inter alia).

[5] existing work in the area, gaps An early example of an equal weight, majority voting-based system in ASR is Recognizer Output Voting Error Reduction (ROVER) (Fiscus, 1997), which aligns multiple systems with eachother to then vote on each aligned word slot, producing an aligned output string reached by model concensus. More dynamic approaches on the same theme have been explored as with Agrawal et al. (2023), whereby a weighting module is trained on transcriptions to select expert models so as to to minimize word error rate (WER), leading to an improvement over equal-weight voting. A number of confidence-based approaches have been taken towards language identification for both HMM and more modern E2E architectures Gitman et al. (2023), whereby only the outputs of the most confident model from the ensemble are used. One issue with deriving reliable confidence measures for modern E2E deep neural network (DNN)-based models lies in their tendency toward overconfidence in ouptput predictions (Wei et al., 2022), though some mitigation strategies have been introduced to derive more reliable confidence estimates from entropy calculations (Laptev and Ginsburg, 2023). Ensembles have been utilized in a number of ways for phone-level MD as well. Ananthakrishnan et al. (2011) investigated MD with a series of binary classifiers to detect mispronunciations typical of Swedish learners of varying L1 backgrounds, focusing on systematic phoneme confusion patterns over several utterances and exeplifying through two case studies how this approach could be used to motivate targeted interventions. Calik et al. (2023) also pursued MD for Arabic phonemes using a variety of model architectures and combination techniques i.e. bagging, boosting, stacking, and majority voting, finding majority voting the most performant.

[6] Contribution of current work Although model ensembling has a long history in ML and ASR, confidence-based ensembled have not to our knowledge been leveraged toward phone-level MD. Despite the relative scarcity of ensembling applications in MD, the limitations of data and model expertise motivating general ensemble applications clearly apply to L2-speech-specific phone classifiers as well: monolingual data to train expert models for Irish and English is available in abundance, but phonetically labeled data for Irish L2 speakers with an English L1 is, like phonetically-annotated L2 data more generally, comparatively scarce. Models tuned to output phonemes for English would not be expected to reliably transcribe Irish-specific phonemes, and models tuned to Irish would similarly underperform for English phonemes. But given the possibility of combining these expert systems in an ensemble, could each constituent expert model's phone coverage be leveraged to complement the other's towards identifying mispronunciations in Irish L2 speech? If combined as a confidence-based ensemble of monolingual expert systems, we might for example expect an English model to more confidently predict English-like phones in an attempted Irish utterance with heavy interference from the speaker's L1, while a more native-like pronunciation for other segments without any L1 interference might be confidently transcribed by the Irish ASR. If we choose the most confidently predicted phone from each frame, we might be able to derive a combined string that more accurately reflects the segmental errors present in learner speech than is possible for each model on its own.

3.2 Data Generation with Synthetic Speech

On the other end of common solutions to L2 data scarcity, we encounter *data augmentation* and *data generation*. Data augmentation is frequently done through labeling of existing audio (see Bartelds et al., 2023; Xu et al., 2020; Yang et al., 2022, inter alia) or altering existing audio through targeted alternations to phonemes or other perturbations (Kheir et al., 2023). Automated labeling of unlabeled audio presents a promising avenue for low-resource languages, as it is easier to collect this kind of data than manually annotated data (Bartelds et al., 2023), and for languages with at least some digital presence, there may be latent reservoirs of this kind of data available to propel such an approach. In ASR generally, Hayashi et al. (2018) pursued back-translation-like data augmentation was demonstrated to bolster performance, whereby a text-to-encoder translation model is trained to predict ASR hidden states, then using these to retrain the decoder without need for any speech paired to the augmenting text data. For MDD, Yang et al. (2022) illuminated the use of momentum pseudo-labeling via a teacher-student model ensemble using Wav2Vec2 as a promising way to leverage plentiful unlabeled L2 speech for improving MDD performance.

In recent years, however, as end-to-end ASR models like Wav2Vec2 become increasingly capable, so to do their TTS counterparts, advancing to the point where the speech they produce is nearly indistinguishable from human speech in naturalness. This development has led a number of researchers to pursue synthetic data generation via TTS as another way to fill the data gap for low-resource ASR (see Bartelds et al., 2023; Fazel et al., 2021; Thai et al., 2019, inter alia). Use of TTS outputs to augment data has been demonstrated to provide improvements in WER for ASR models (Bartelds et al., 2023), with the caveats that not all languages have readily available TTS systems. For MDD, however, this inequality in TTS availability is less of an obstacle to providing useful data. The task of generating mispronunciation data has been framed in recent years by Korzekwa et al. (2022) and D. Zhang et al. (2022) as *phoneme paraphrasing*, as an analogue to the paraphrasing task in Natural Language Generation (NLG). The former approach by Korzekwa et al. (2022) proposes three avenues for synthesis—phoneme-to-phoneme (P2P), TTS, and sequence-to-sequence (Seq2Seq)—all improving accuracy in detecting mispronunciation errors. The latter approach focuses on a P2P pipeline, where canonical phone strings are translated to L2-like variants using a Seq2Seq MT model, which is then diversified using their diversified and preference-aware decoding (DPAD) module to improve generalizability, and finally fed to a TTS system L1 to generate L2-like mispronounced audio. Critically, a seed set for training the Seq2Seq paraphraser was extracted from the expertly-annotated dataset used L2-arctic, resources not necessarily available in our case for Irisih. Similarly, the preference term enriching the variety of their DPAD system relies on curated sets of mispronunciation patterns specific to L1 Mandarin speakers.

Similar to D. Zhang et al. (2022) we see the potential for synthetic data as a possible answer to MD data scarcity through Seq2Seq translation to create controllable errors mirroring what one would expect from language learners with the same L1 the TTS is tailored to. Mapping phonetic transcriptions of canonical Ulster pronunciations to inputs acceptable to an English TTS gives highly English-sounding approximations of Irish speech with exactly the kinds of segmental errors one expects from EnglishL1 learners of Irish[1]. Instead of using a NN-based architecture for our phone-to-phone paraphrasing, we use a monolingual English ASR model fed with Irish speech for which we have corresponding phonetic transcriptions. As we do not have an expertly annotated seed corpus to draw from with Irish as their work does by drawing the seed data from L2-arctic, we considered a zero-shot approach that circumvents the need for expensively annotated L2 data desireable, an approach which we will elaborate on in the following chapter. L2-gen provides greater diversity through their DPAD augmentation system, but in the interest of time, we do not attempt here to introduce additional variance to the TTS voices. However, we recognize this innovation as a critically important one for promoting more robust performance and will explore possible avenues for mimicking this behavior with an ASR-based system in the Discussion section.

[1]Another noteworthy property of the recordings is the English-like prosody and stress patterns, which could be pursued in an investigation into suprasegmental deviation from Ulster speech, but lies outside the scope of the current work.

4 Methods & Materials

In the following chapter, we will elaborate the core experiments explored for the current work: one which explores ensembling monolingual models as described above as a solution to MDD in the face of data scarcity; the other explores data augmentation via an English ASR model transcribing TTS inputs from Irish speech to create the mispronounced data required to train an ASR model directly for MDD. We will begin with detailing the features in common between these experiments before continuing to each experiment in turn, and end with the evaluation details shared between experiments.

4.1 General Experiment Setup

At a high level, both of the MD pipelines explored follow the same general flow, starting with audio input of L2 speakers speaking Irish. The input waveform is then processed by one or several ASR models, configured to output strings of phonemes. These phoneme strings are then aligned to canonical IPA transcriptions of the target string taken from a phonetic grapheme-to-phoneme (G2P) model, with corresponding audio scraped from the online Irish pronunciation dictionary Teanglann, to be fully described in the data section Teanglann. Manually annotated IPA strings serve as our ground truth labels, which are also aligned to the canonical string, and then scored for mispronunciation, a procedure we will expand upon in the Evaluation section.

The model chosen for all experiments was the 300 million parameter version of Wav2vec2 XLS-R (Babu et al., 2021). This choice of architecture was motivated by the desire to leverage the aforementioned pre-training/fine-tuning paradigm on a model shown by Babu et al. (2021) to improve on previous work most notably for low- and mid-resource languages. Selection of the 300 million parameter version of this architecture was primarily motivated by desire to rapidly train models and test ideas, aiming for proof-of-concept over beating the state-of-the-art. These models were fine-tuned on phonetically annotated datasets, allowing us to obtain phoneme-level transcriptions for comparison.

All models were implemented using the Hugging Face *Transformers* library (Wolf et al., 2020). Training was managed with the Transformer *Trainer* class, configured via *TrainingArguments*. Training progress and metrics were logged to Weights & Biases(and others Biewald, 2020). Our implementation of Wav2Vec2 for phoneme output followed the publicly available tutorial by Vitouphy on Kaggle[1]. Models were instantiated with an attention dropout of 0.1, layer dropout of 0, feature projection dropout of 0, mask time probability of 0.75, mask time length of 10, mask feature probability of 0.25, and mask feature length of 64.

Training for all ASR models used identical parameters, with batch size of four, learning rate 3e-5, 2000 warmup steps, and 16-bit precision training to reduce memory usage and speed up training. Minimizing character error rate (CER) was used as the evaluation metric and in guiding early stopping, which had a patience of 6 epochs

[1] https://www.kaggle.com/code/vitouphy/phoneme-recognition-with-wav2vec2

with a maximum of 50 epochs of training to avoid overfitting. Training was carried out on kaggle[2] using their P100 GPU. Full configuration details are given in Table 4.1.

Table 4.1: Model architecture and training configuration shared across all experiments.

Category	Parameter	Value
Architecture	Base model	XLS-R 300M
	Attention dropout	0.1
	Layer dropout	0.0
	Feature projection dropout	0.0
	Mask time probability	0.75
	Mask time length	10
	Mask feature probability	0.25
	Mask feature length	64
	CTC loss reduction	mean
Training	Batch size	8
	Gradient Accumulation Steps	4
	Learning rate	3e-5
	Evaluation strategy	Epoch
	Learning Rate warmup steps	2,000
	Learning Rate scheduler	linear
	Precision	16-bit
	Objective	CER
	Early stopping patience	6 epochs
	Max epochs	50
	Optimizer	Adam

4.2 Experiment 1: Monolingual Model Ensembling

In this section we will begin by describing the data used, first for the English model, then the Irish model. Following this we touch briefly on the preprocessing procedure before expanding on the full experimental setup.

4.2.1 Data

To train our monolingual systems, we procured two monolingual corpora of read speech audio with phonetic transcriptions. Our English model was trained on the TIMIT corpus Garofolo et al. (1993) with the Irish model trained on audio data from the online Irish Dictionary and Language Library[3].

TIMIT

The TIMIT corpus is a read speech corpus of American English speakers who spent their childhoods in one of eight major dialect areas of the United States. These areas are widely recognized with the exception of the Western dialect region, where boundaries are not confidently delineated, and the "army brat" group, consisting of speakers which frequently moved during their childhood due to the demands of highly mobile military service member parents, resulting in exposure to a variety of dialects (Garofolo et al., 1993). The full dataset consists of 6300 sentences from 630 speakers, so ten

[2]https://www.kaggle.com
[3]https://www.teanglann.ie/en/

sentences per speaker. The duration of the data loaded from the Kaggle dataset linked to the TIMIT github dataset[4] lies at two hours and forty minutes. Note that this is not the full dataset, but to ensure some level of comparability between the monolingual models so as to not advantage one over the other, this data was left deliberately downsampled. The TIMIT phonetic transcriptions were given in a revised version of ARPABET, a set of transcription codes developed by Advanced Research Projects Agency (ARPA)(Rice, 1976, for the original set, see).

Teanglann

The Dictionary and Language Library is an online resource developed by Foras na Gaeilge[5] in conjunction with the New English-Irish Dictionary. Among the resources available are The Pronunciation Database, which contains recordings of individual words spoken by native speakers from three major dialects: Connacht, Ulster, and Munster. The Ulster recordings used for our monolingual Irish model were acquired via webscraping adhering to the site's robots.txt file limit of one request per two seconds. The words scraped were drawn from an available G2P dictionary to ensure scraped audio had a corresponding canonical IPA transcriptions[6]. The resultant combined dataset had an audio duration of roughly five hours of audio.

Preprocessing

The audio from both Teanglann and TIMIT was loaded and resampled to 16kHz for compatability with Wav2vec2. For the phonetic trancriptions, TIMITs phonetic representations were translated to IPA using the phonecodes library, and maintaining one space between each phoneme or dipthongs[7]. The Irish data was transcribed through a lookup in the aforementioned G2P dictionary, which adhered to the same spacing rules as for TIMIT. For the Irish set, phones from the g2p canonical forms had all but palatal diacritics collapsed to the base phoneme to minimize the label space while preserving this particularly salient phonemic distinction from Irish phonology (see Section 2.1). We will return to this choice in the Chapter 6

We split the datasets into train/dev/test splits with .8/.1/.1 ratios respectively, which were then trained according to the procedure described in Section 4.1. One critical limitation with the Irish data was that due to the acquisition method, we were left with no way verifiable way to identify speakers, so the different splits likely contains speaker overlap. We sought to mirror this limitation with the English data so as to not bias any particular language when ensembling them, and so the monolingual evaluation metrics are also likely overly optimistic.

4.2.2 Experimental Setup

The two core models in the ensemble were Irish and English monolingual ensembles trained on the datasets described above. To combine their outputs, we first derived canonical spans from the torch audio forced alignment library[8] using the Irish model and the canonical transcriptions of the input. Since the output spans using this function on subword units spans only a few frames, we used these output frames as the

[4]https://github.com/philipperemy/timit

[5]Foras na Gaeilge is a group which promotes the Irish language, supports Irish-medium education, and advises public and private sector organizations, among its other functions. https://www.forasnagaeilge.ie/

[6]The dictionary of words to scrape was derived from an Ulster Irish G2P file generously provided by Jim O' Regan

[7]dipthongs were kept together as a single single unit to keep with standard practice

[8]torchaudio forced_align documentation

middle point to expand the canonical spans until they met a neighboring span. At this point we found each model's most confident phoneme over each span along with its corresponding confidence value, and compared their confidence to determine which model's output we would use for each span's slot in the final decoded string. The baseline confidence value used was raw logit probability, but we tested confidence values calculated from gibbs entropy following Laptev and Ginsburg (2023) and Gitman et al. (2023) as seen in 'Experimental Setup' on page 25. Aside from raw confidence comparison we tested the use of an Logistic Regression (LR) model selection block following Gitman et al. (2023) to mediate the confidence values in case one model was systematically more confident than the other. This LR block was trained on 99 manually annotated word-level utterances held out from the evaluation data described in more detail at the end of the chapter.

$$\mathrm{F}_g^e(p) = \frac{V \cdot e(\sum p_v ln(p_v)) - 1}{V - 1} \tag{4.1}$$

Due to the relatively granular distinctions separating some Irish phonemes, particularly palatalized consonants, we expected confidence values to be systematically lower for the Irish model for these palatalized/velarized pairs than English would in discriminating the class as a whole. One solution explored to overcome this issue was to add a Russian model to the ensemble as a kind of high-resource palatalization authority, since the language shares a considerable number of palatalized phonemes with Irish. In the ensemble procedure described above, the Russian model outputs, when included, were chosen automatically if the output phoneme was palatalized. This model[9] was a Wav2Vec2-large-xlsr-53 variant taken from the Huggingface Model Hub and was fine-tuned on the commonvoice Russian dataset to output phonemes, which could be readily mapped to IPA by mapping dictionary included with the model.

Another potential solution to the problem of diluted model confidence was explored in simply pooling the values of related phonemes (e.g. /l_{vl}/ & /l_{pal}/) when choosing between the models, and then the most probable phoneme among those if the Irish model was chosen. The intuition guiding this can be visualized below4.1.

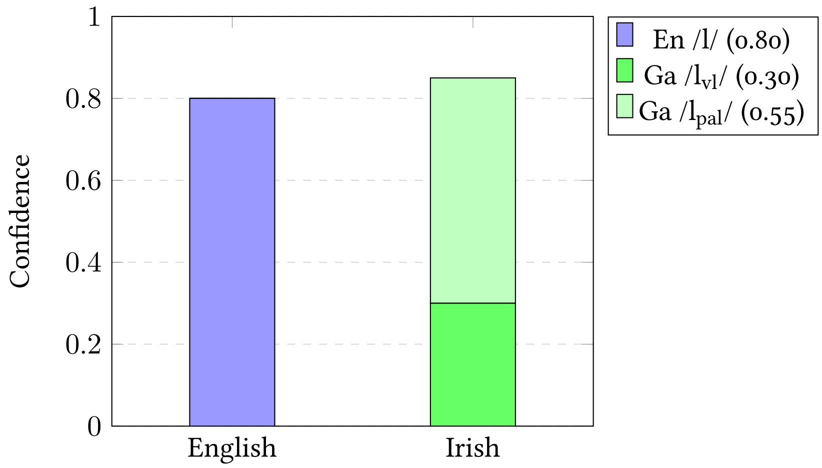

Figure 4.1: Phoneme pooling for a span with canonical /lʲ/. Without pooling, the Irish model's split distribution would give it lower confidence than English and lose. Summing probability mass across the broad/slender family (shaded, 0.85 total) beats English (0.80), and the Irish model's top prediction /lʲ/ is selected.

[9] HuggingFace model card

4.3 Experiment 2: Synthetic Mispronunciation Data

In this section we will confine ourselves to describing the how the synthetic data was derived and processed into a training set. The model itself was trained on this set identically to the procedure described at the beginning of the chapter. Decoding functions identically to the ensemble, whereby the Irish monolingual model is used to force align the canonical string, defining the spans from which we search for the most probable output phoneme.

4.3.1 Data

Canonical recordings were obtained from the Teanglann pronunciation dictionary, with corresponding canonical IPA transcriptions from the aforementioned Ulster Irish G2P dictionary: the same dataset as that used for training the Irish model in Experiment4.2. This canonical data was complemented with augmenting learner-like recordings obtained via Google Cloud text-to-speech(Google Cloud, 2026) using the full set of American wavnet(van den Oord et al., 2016) voices at a recording frequency of 16kHz. Unlike the Seq2Seq phoneme paraphrasing model used by D. Zhang et al. (2022), we opted for a simple ASR approach to acquiring Speech Synthesis Markup Language (SSML) inputs. To do this, we fed our canonical Irish recordings through the monolingual English model described above, whereby we obtained transcriptions of realistic English-L1 mispronunciations of the Irish speech. The input was given in SSML to enable direct use of IPA transcriptions to guide pronunciation.

These mispronounced synthetic samples together with the canonical Irish recordings of each word composed the *positive* and *negative* utterance pairs of the training data, similar to the *adversarial sampling* strategy used by D. Zhang et al. (2022) but in our case, only with one adversarial utterance in the training sample. The transcriptions and recordings of each pair were concatenated with ordering of the language randomized.

4.4 Evaluation

In the following section we detail the evaluation data used for evaluating each experiment, as well as the process used to annotate it. Following this, we describe our evaluation procedure: the metrics and statistical test used. include table of models with evaluation set performance

4.4.1 Data

To measure an MDD system's sensitivity to plausible Irish L2 errors, we first needed a source Irish L2 speech. We began with the Irish portion of the Few-shot Learning Evaluation of Universal Representations of Speech (FLEURS) dataset(Conneau et al., 2022). The crowdsourced nature of the dataset allowed for speakers of various skill levels to contribute, an attribute that is perhaps not attractive for most ASR applications, but which perfectly suits our purpose, as there were some markedly non-native speakers in the evaluation set. Canonical transcriptions of the data was derived from the aforementioned G2P dictionary, with a fallback lookup for out of vocabulary (OOV) consisting of an Montreal Forced Aligner (MFA) G2P model also trained on this dictionary. The minority of sentences left with unknown tokens were removed, which largely stemmed from proper nouns consisting of letters outside of what the G2P fallback could handle. Audio volume was normalized with ffmpeg(FFmpeg Team, 2024), specifically ffmpeg-normalize at default settings. White-noise handling was carried out with ffmpeg Adaptive Fast Fourier Transform Denoise with a noise floor of -50 dB, white noise for noise type, and otherwise default settings.

To prepare the FLEURS dataset for use in phoneme-level MDD, a small subset of the audio data was manually segmented and transcribed in IPA to establish a ground truth set of transcriptions using a Label Studio html interface created by the author (an example of which can be seen below in Figure 4.2) to transfer data to different machines via Google Cloud Service (GCS). Of the audio segments produced in this way, 99 were used as training data for the aforementioned LR selector block in Section 4.3, and the remaining 146 were used for evaluating both experiments. The process of capturing these deviations consisted of the following steps: the audio was loaded into an interface with the canonical IPA transcriptions as well as orthographic representations included in the interface; recordings were then repeatedly assessed for natural breaks in speech that could be used to anchor clear segments to; these segments were then repeatedly analyzed for perceived realization within the bounds of valid English or Irish phonemes used for model training, referring to interactive the interactive ipachart.com when unsure about the sounds produced. This process was carried out by the author only, a non-native Irish speaker of modest ability, with no professional phonetician training, and with limited reference material to refine my annotations on with respect to palatalized/velarized consonants, all notable limitations we will return to later. The choice to have the target canonical string visible during annotation had a number of unintended negative effects on the annotation process that will also be discussed in the 'Discussion' on page 33 chapter.

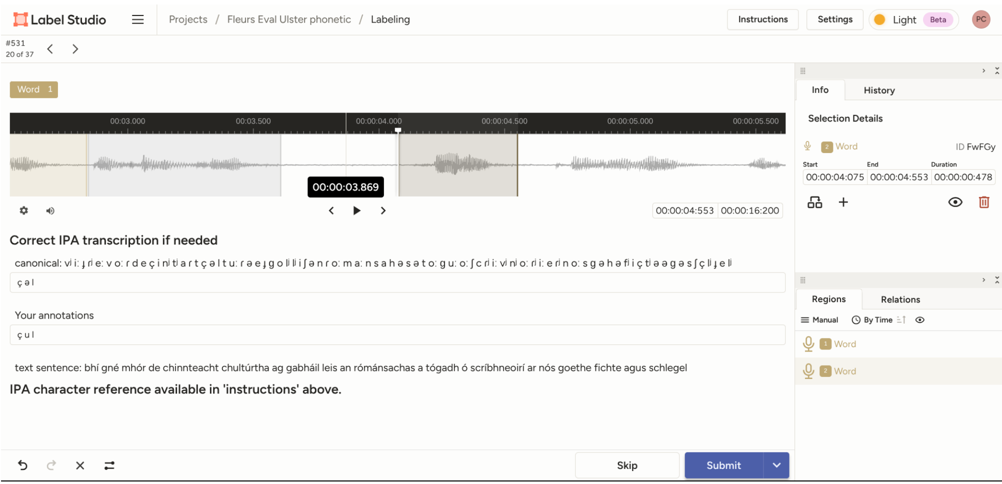

Figure 4.2: Annotation interface created in Label Studio to assist in phonetic annotations. Annotator manually defines segments, drags the canonical representation of the segment to the canonical slot, and annotates perceived production in the ground truth slot.

4.4.2 Evaluation procedure

We evaluate phone-level discrimination of the ASR by first aligning canonical strings to the ground truth annotated strings with the Needleman-Wunsch algorithm (Needleman and Wunsch, 1970) with the PanPhon(Mortensen et al., n.d.) library in place of uniform mismatch costs to help better align sequences with articulatorily similar phonemes. Alignment of the model outputs and canonical strings are trivial as they are the same length by definition as a consequence of the canonical string defining model decoding spans for each experiment through False Acceptance (FA). This alignment by architecture then trivializes the alignment of model outputs to corresponding

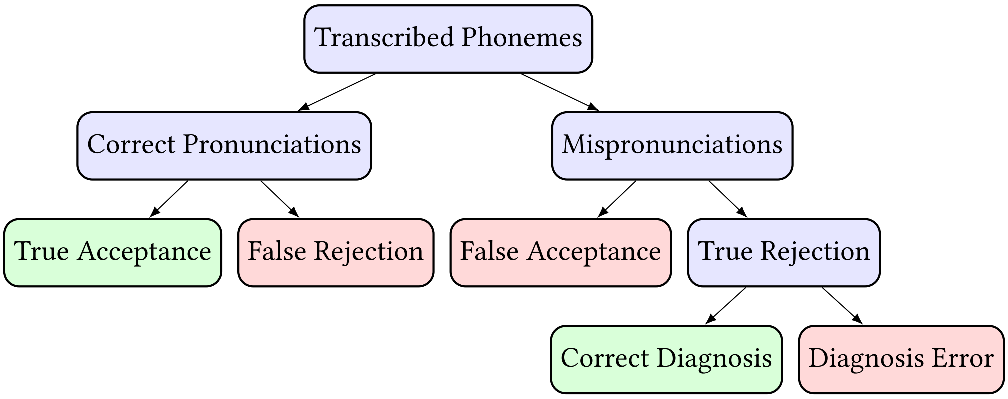

Figure 4.3: General Evaluation hierarchy of phoneme level MDD. The lowest level on the right is where diagnosis happens for correctly rejected phonemes: were the phonemes detected and rejected by the system actually those present in the mispronounced ground truth labels?

ground truth strings, as they are equal to the alignment of the canonical string to the ground truth string. This same feature renders the whole decoding system architecturally blind to insertions however, as models can not decode a phoneme outside the force aligned spans—an issue we will discuss in 'Discussion' on page 33.

4.4.3 Metrics

For *detection*, we can focus on alignment between canonical and ground truth to True Acceptance (TA) True Rejection (TR), False Rejection (FR) and FA. At each canonical phoneme position, the correctly pronounced/mispronounced verdict is given by the canonical/ground truth alignment. Then, given the canonical/model hypothesis alignment, TA refers to those also determined to be correct by the model, while FR refers to those which are deemed incorrect by the model but correct by the canonical/ground truth. Among the pronunciations deemed *mispronounced* by the canonical/ground truth alignment, those accepted by the model/canonical alignment are considered correctly pronounced as FAs, while those considered mispronounced are accurately rejected as TR. At the level of diagnosis, we are interested in further analysis of those pronunciations which fall under TR, specifically for our purposes of segmental pronunciation analysis, for each TR, is each phoneme detected by the ASR model in agreement with the aligned ground truth string? Those that are are labeled with Correct Diagnostic (CD), and those which are not, Diagnostic Error (DE). An illustration of this evaluation hierarchy is available in Section 4.4.3.

In line with prior MDD research(Leung et al., 2019; K. Li et al., 2017; Qian et al., 2010, inter alia), we focus on Precision, Recall, F1-score, False Rejection Rate (FRR), False Acceptance Rate (FAR), and Diagnostic Error Rate (DER) to evaluate performance. False Rejection Rate (FRR), False Acceptance Rate (FAR), and Diagnostic Error Rate (DER) are calculated from the values above in Equations (4.2a) to (4.2c), and Precision, Recall, and F1-score are calculated in Equations (4.2d) to (4.2f).

$$FRR = \frac{FR}{TA + FR} \quad (4.2a) \qquad FAR = \frac{FA}{FA + TR} \quad (4.2b) \qquad DER = \frac{DE}{CD + DE} \quad (4.2c)$$

$$P = \frac{TR}{TR + FR} \quad (4.2d) \qquad R = \frac{TR}{TR + FA} \quad (4.2e) \qquad F1 = \frac{2 \cdot P \cdot R}{P + R} \quad (4.2f)$$

For statistical testing, we utilized McNemar's test for classifier comparisons from the mlxtend library(Raschka, 2018). This test can be used for machine learning to compare predictive accuracy of two classifiers(Dietterich, 1998). For this test, we evaluate each model on a test set and count the instances where they were correct or incorrect and construct a contingency table with 4 cells corresponding to each combination of model classifications, the relevant values being b and c, or the off-diagonal disagreement counts between models compared.Table 4.2

	Baseline correct	**Baseline incorrect**
Model correct	a	b
Model incorrect	c	d

Table 4.2: McNemar contingency table for paired significance testing of two systems evaluated on the same set of phone positions. Each cell counts positions where the respective combination of outcomes occurred. Only the discordant cells matter for the test: b (system correct, baseline wrong) and c (system wrong, baseline correct).

For the null hypothesis to hold, neither model performs better than the other, which formally translates to the probabilities p(b) and p(c) from Table 4.2 being the same. The test statistic can be calculated as seen in Equation (4.3) and can, after setting a significance threshold $\alpha = 0.05$, be used to calculate a p-value, or the probability of observing the counts b and c under the null hypothesis $p(b) = p(c)$. If the p-value lies under the significance threshold, we can reject the null hypothesis.

$$ {}^2 = \frac{(b-c)^2}{(b+c)} \tag{4.3} $$

5 Results

In this chapter we will present the results of the the above experiments. We will first report the evaluation performance of each model on its evaluation dataset in terms of CER, continue with the synthetic experiments, then continue on to the ensemble experiments, before presenting the most performant model from each experiment alongside the baseline. Finally, we will present each ablations performance with respect to the full version of each experiment in terms of F1 percentage difference.

Table 5.1: Constituent model evaluation results

system	Eval CER
Irish monolingual model	0.167
English monolingual model	0.124
Multilingual ablation model	0.161
Synthetic ablation model	0.112
Full Synthetic model	0.089

For the synthetic experiments, the most performant model by way of F1 score was the unpaired ablation, though the model trained with contrastive paired samples did better by way of FAR and DER. Despite this, given our $\alpha = 0.05$ significance cut-off, no model outperformed baseline to a statistically significant degree, which one can see in Table 5.2.

remember to bold highest values in column

Table 5.2: Mispronunciation detection results: synthesis experiment.

system	Prec.	Recall	F1	FAR	FRR	DER	McNemar
Irish only baseline	0.444	0.756	0.559	0.244	0.342	0.687	—
L2 model unpaired	0.469	0.760	0.580	0.240	0.310	0.734	0.053
L2 model paired	0.431	0.809	0.562	0.191	0.386	0.660	0.282

For the Ensemble experiments, the most performant model by way of F1 score is again the naive ablation model, in this experiment trained on a combined set of English and Irish. The 2-way ensemble with a probability confidence function was the most performant among models without a selector block as seen in Table 5.3. However, even the least performant of the selector block models outperformed it. The most performant selector block model was the 2-way model with the gibbs confidence function and pooled confidence values. All models did not perform above baseline, with all but the ablation model underperforming baseline to a statistically significant degree as seen in Table 5.4

For the main results, one can see how the baseline compared against both naive ablation models, the paired synthetic model, and the most performant ensembles with and without a selector block in Table 5.5.

For relative ablation performance, we can see for the synthetic experiment that the removal of adversarial sampling improved F1 by 3.17 percent compared to the full model in Table 5.6. For the ensemble, the removal of the selector block reduced F1

Table 5.3: Mispronunciation detection results: ensemble ablations (no selector).

system	Prec.	Recall	F1	FAR	FRR	DER	McNemar
Irish only baseline	0.444	0.756	0.559	0.244	0.342	0.687	—
Irish+English	0.423	0.794	0.552	0.206	0.392	0.678	0.101
2-way, gibbs	0.362	0.859	0.510	0.141	0.546	0.662	0.000*
2-way, gibbs+pool	0.367	0.844	0.512	0.156	0.526	0.719	0.000*
2-way, prob	0.366	0.870	0.515	0.130	0.545	0.654	0.000*
2-way, prob+pool	0.366	0.847	0.511	0.153	0.531	0.712	0.000*
3-way, gibbs	0.362	0.859	0.509	0.141	0.548	0.662	0.000*
3-way, gibbs+pool	0.367	0.844	0.511	0.156	0.527	0.719	0.000*
3-way, prob	0.365	0.870	0.515	0.130	0.546	0.654	0.000*
3-way, prob+pool	0.365	0.847	0.510	0.153	0.532	0.712	0.000*

Table 5.4: Mispronunciation detection results: ensemble ablations (with selector).

system	Prec.	Recall	F1	FAR	FRR	DER	McNemar
Irish only baseline	0.444	0.756	0.559	0.244	0.342	0.687	—
Irish+English	0.423	0.794	0.552	0.206	0.392	0.678	0.101
2-way, sel/prob	0.376	0.832	0.518	0.168	0.499	0.638	0.000*
2-way, sel/prob+pool	0.380	0.824	0.520	0.176	0.487	0.676	0.000*
2-way, sel/gibbs	0.377	0.828	0.519	0.172	0.494	0.641	0.000*
2-way, sel/gibbs+pool	0.386	0.821	0.525	0.179	0.472	0.693	0.000*
3-way, sel/prob	0.375	0.832	0.517	0.168	0.501	0.638	0.000*
3-way, sel/prob+pool	0.379	0.824	0.519	0.176	0.488	0.676	0.000*
3-way, sel/gibbs	0.377	0.828	0.518	0.172	0.495	0.641	0.000*
3-way, sel/gibbs+pool	0.385	0.821	0.524	0.179	0.473	0.693	0.000*

the most by 2.56 percent. Removal of pooling reduced F1 by 1.24 percent, replacing Gibbs confidence with raw probability reduced F1 by 0.98 percent, and removal of the Russian model instead improved performance by 0.12 percent as seen in Table 5.7.

Table 5.5: Mispronunciation detection results: main comparison.

system	Prec.	Recall	F1	FAR	FRR	DER	McNemar
Irish only baseline	0.444	0.756	0.559	0.244	0.342	0.687	—
L2 model unpaired	0.469	0.760	0.580	0.240	0.310	0.734	0.053
L2 model paired	0.431	0.809	0.562	0.191	0.386	0.660	0.282
Irish+English	0.423	0.794	0.552	0.206	0.392	0.678	0.101
2-way, prob	0.366	0.870	0.515	0.130	0.545	0.654	0.000*
2-way, sel/gibbs+pool	0.386	0.821	0.525	0.179	0.472	0.693	0.000*

Table 5.6: Ablation study: synthesis experiment.

system	F1	Δ (%)
Full (L2 model (paired))	0.562	—
– adversarial sampling	0.580	+3.17

Table 5.7: Ablation study: ensemble experiment.

system	F1	Δ (%)
Full (Ensemble (3-way, sel/gibbs+pooled))	0.524	—
– selector	0.511	-2.56
– Russian	0.525	+0.12
– pooling	0.518	-1.24
– gibbs	0.519	-0.98

6 Discussion

As shown above, the synthetic-augmented models and multilingual model with Irish plus English data did not produce significant improvements above baseline, and the ensembles significantly underperformed baseline. In this chapter, we will begin by exploring differences in relative performance between ablations within each experiment to examine componental contributions to performance. Following this, we will explore confounds that could have contributed to overall underperformance of the present experiments, which we will consider from implementational to structural in nature.

6.1 Experiment 1: Model Ensembling

The ensemble results also show some relative behavior between systems worth digging into. The selector block contributed the most to full system performance, but it is noteworthy that using Gibbs over raw output probability resulted in *decreased* F1 for the ensembles without a selector block, while *increasing* F1 for those with a selector block, especially when paired with pooling. This seems to correlate with the relative contributions of the English and Irish models to the ensemble with different confidence calculations, with or without a selector, and with pooling added to the combination as seen in Figure 6.1. The same components which increase performance also seem to balance the selection between models, in particular the selector block. That the selector block serves to balance the model selection comes as no surprise given that was its intended function, but it is worth reiterating its contribution to performance in this context.

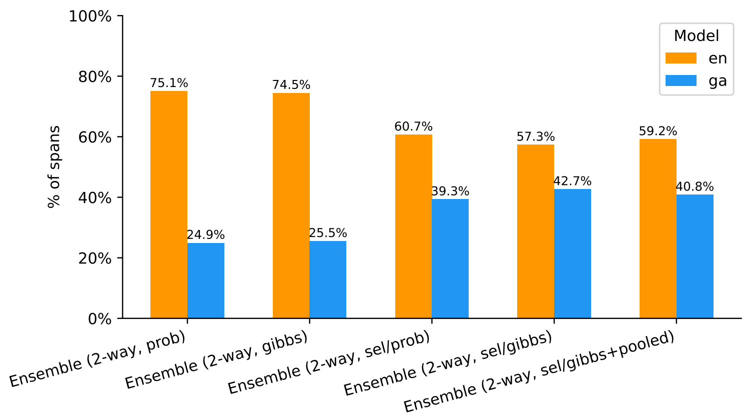

Figure 6.1: Ensemble winner contributions, en/ga

Removing the russian model improved performance slighly, but it is worth remembering that the russian role in the ensemble was as a palatalization specialist, and in the most aggressively implemented variant possible: if the russian model detected a palatalized phoneme, that was what got added to the final string without voting between the other ensemble members. As a more equal member of the ensemble that participates in voting, perhaps by combining its palatalization evaluation with Irish or contributing towards majority voting, it might contribute to performance rather than it hinder it. However, if one compares the palatalizations detected by the russian inclusion in its most aggressive form against those detected by the pooling strategy, one sees that they are solving the same problem about 82.4% of the time in Figure 6.2, with russian inclusion detecting slightly more than the pooling strategy when alone, but also contributing with some false positives as we can see due to its relatively lower performance. This suggests pooling might be a lighter weight alternative to a specialist model in boosting palatalization detection in an ensemble configuration. Since every model addition increases memory use and inference time, compared to a relatively simple mapping table with pooling, this trade-off might be worth considering closely.

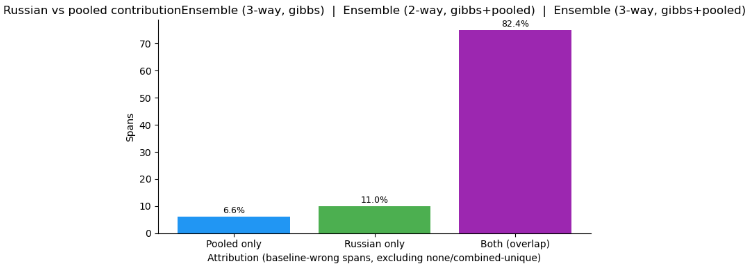

Figure 6.2: Relative performance in palatalization detection between Russian inclusion, pooling, and both in percentage terms.

Although ensemble components improved on the simplest version of the ensemble, the multilingual ablation and monolingual baseline still outperformed all ensemble versions. This suggests that a multilingual model might be a better starting point for combining monolingual data than an ensemble of monolingual models.

6.2 Experiment 2: Synthetic Mispronunciation Data

Beginning with the synthetic ablation, we can see that the removal of adversarial sampling (i.e. concatenation of canonical and synthetic pairs of speech and text) improved performance relative to the full model. Training instances for this ablation were thus single word utterances, which matched the evaluation data segments also consisting primarily of single word utterances. Although this component was shown to contribute the least to the out-of-domain experiment by D. Zhang et al. (2022), it is worth considering that the mismatch between training and evaluation format for the full model may have degraded performance relative to the ablation in our experiment despite outperforming it when evaluating the models on data derived from the same source as their respective training data. Given these limitations, due caution should be exercised in discounting it's potential contribution in other settings.

Another possible explanation for system underperformance could lie in the absence of a TTS diversification mechanism similar to DPAD in D. Zhang et al. (2022). Intuitively, only capturing one mispronunciation variant does not give a full sense of variation for mispronunciations of a word. Augmenting our model data with a fuller range of possible mispronunciations for each word should bolster performance and we will discuss a possible adaption of this concept to our ASR-based approach in Chapter 8.

6.3 Implementational Confounds

Whatever might account for relative differences between models, their uniform underperformance with respect to baseline calls for a closer examination of possible causes common to both experiments. Firstly, our decision to decode strings based on canonical-anchored Force Alignment spans for all models was not a decision without trade-offs. Pinning spans to the baseline Irish model advantages the baseline since all decoding is performed based on where the baseline decides the spans are. This imperfect decoding strategy was initially intended to address the ensemble problem where aligning free-phone decoded strings with basic mismatch distance led to implausible string alignments that spuriously aligned the tails of one string with the head of another, and heavily punished alignment of phonetically close characters differing only by a few features. This was especially problematic for characters of particular interest like palatalized consonants, which would not reliably match with each other.

One way to solve this problem, and the one we chose here, is to define spans across the audio via Forced Alignment and compare ASR outputs within each span. Aligning multiple ASR model spans was itself not straight forward. CTC trained models exhibit 'peakiness' where spans of individual characters are typically confined to one or two frames (See Zeyer et al., 2021, for a fuller description). Where these peaks occur has little consistency between models. In some cases, they would come at the onset of a sound, other times the end, and some times somewhere in between. Determining reliable phone spans from these frames in CTC models is a well known problem, and our chosen heuristic was to define the canonical frames as middle anchors from where the span was expanded outward until it encountered a neighboring span or the end of the string. Our strategy produced plausible spans and output strings, but also renders the system blind to insertions; if the speaker inserts sounds between where the canonically anchored spans expect only one, the system chooses the most confident candidate and discards the rest. This shortcoming also illuminated a particular shortcoming in the aforementioned decision to collapse the Irish phone set to phonemes stripped of all diacritics except palatalization markers. Some word-initial velar consonants were annotated as simple velar consonants in the Irish G2P model, but are realized with a velar offglide that the ensemble and synthetic models could detect and labelled as a separate phoneme. This was likely due to the English model not collapsing the velar offglide together with the preceding phoneme as the Irish model was trained to do. One can clearly see the implications in this point of failure combined with insertion blindness in Figure 6.3. The limitations above point to a number of failure points worth

Ensemble (2-way, prob) | 10266820967862372597_l5ruv

canonical	f	i	n^j	u:	—
gold	f	i	n^j	u:	v^j
asr	f	w	n	v^j	—

Figure 6.3: Ensemble output where velar offglide /w/ edges out /i/ and /v^j/ edges out /u:/ due to canonical anchoring and insertion blindness.

addressing. If we are to stick with canonically-anchored Forced Alignment spans for our evaluation, determining spans heuristically from CTC model peaks is likely not the best approach. It would likely be beneficial to use MFA models to provide more reliable phoneme spans to be mapped onto the frames of the CTC model from which we can then extract phonemes. Insertion blindness is not a necessary consequence of

Forced Alignment methods, rather an implementational compromise that simplified aligning the ASR outputs to the canonical string. One way around this might be to do a version of free-phone decoding where we align the string outputs in a span to the canonical string, but at that point, we are most of the way to free-phone decoding and string alignment. Pursuing this might be worth considering if using feature distances from libraries like PanPhon in the place of uniform mismatch penalties for alignment, as this is a much conceptually simpler approach that handles granular distinctions like Irish palatal/velar contrasts with minimized mismatch penalties: precisely the problem that motivated Forced Alignment span anchoring in the first place. We already align the ground-truth string to the canonical in this way, and switching the ensemble alignment to this strategy would bring it conceptually closer to classic ROVER configurations whereby one progressively aligns constituent models to eachother into a Word Transition Network. One final limitation worth reflecting on is how to handle tokenizer mismatch such as between Irish encoding velar offglides as one token vs English treating them as two. This could be solved with a simple mapping of velar or palatal offglides to the nearest Irish equivalent at evaluation, but ensuring annotation comparability is a problem worth considering carefully.

6.4 Structural Limitations

6.4.1 Annotation Reliability

It is worth reflecting on what a lone annotator who is not a native speaker of the Irish can realistically hope to achieve in annotating the language for evaluation data. First

6.4.2 Cross-corpus comparability

6.5 Final Thoughts

Given the possibility that these attempts were implemented correctly as intended and evaluated properly under the data constraints we were obliged to contend with, we cannot fully distinguish the efficacy of the methods from the ability of the evaluation to measure them. We concede that there are implementation limitations that could be improved and perhaps lead to better performance, but also see reason for caution with respect to evaluation robustness. To reiterate, this is not to say that these methods would have succeeded under more charitable constraints, simply that the experiment as presented here cannot settle the question.

Because the structural limitations on the evaluation impose limits on whether an improvement can be cleanly detected, the most pressing priority would be to establish a reliable evaluation infrastructure, taking into consideration reliability and comparability among datasets used. If that condition is fulfilled, we can consider technical extensions potentially worth pursuing.

[7] limitations belong here too

[8] placeholder: discriminative training improve over standard procedure (Qian et al., 2010) maybe use this for synthetic data to train using the synthetic as mispronounced?

[9] placeholder: annotation limitations

[10] placeholder: check avg confidence distributions per model, is LR necessary?

[11] placeholder: highlight difference in LR function between my and gitman's implementation. check phoneme confidence values between models: if they differ meaningfully, perhaps more advanced LR formulation is needed.

[12] placeholder: constituent model confidence values

Deichler et al. (2024) multimodal conversational dataset with cospeech gestures. move to discussion J. Zhang et al. (2021) open source speech corpus speech ocean for pronunciation assessment Zhao et al. (2018) L2 arctic dataset

[13] Summarize the main findings from results and how it relates to the research questions (Conneau et al., 2020) low-resource languages benefit more from similar higher-resource languages.

[14] why confidence might need adjusting? (Laptev and Ginsburg, 2023)

[15] Detail possible applications to Language learning, the role of results in augmenting self-directed language learning Hardison (2005) effect of multimodal input on speech identification

[16] Detail possible (or actual if time allows) Furhat application Deichler et al. (2024) multimedia data

[17] touch on improved contrastive training as possible way forward Khosla et al. (2021) contrastive learning with labeled data.

[18] Argue for accurate articulator representation in digital agents as supportive pillar in pronunciation feedback. Rosenblum (2008) speech perception as multimodal phenomenom

7 Conclusions

In this paper we explored two avenues of overcoming data-sparcity for CAPT in the low-resource conditions faced by Irish. Neither the models trained on synthetically augmented data nor the multilingual ensembles provided a statistically significant improvement over baseline, with all versions of the ensemble significantly underperforming it.

There are a number of limitations worth mentioning, however. Foremost among them perhaps being annotation reliability and cross-corpus comparability. continue here

In light of these caveats, we do not believe it is possible to draw strong conclusions about the presented results, neither positive nor negative. Though the methods used draw upon works that have shown more encouraging performance when applied under less challenging data constraints, there is no guarantee these approaches are suitable for MDD, but it may be worth testing them under more charitable constraints regardless. There are conceptually similar approaches used in other ASR domains which might be worth extending to address data constraints common to MDD, but the results seen in the current work reiterate the need for a minimum of reliable evaluation data.

[19] reiterate how results connect with research questions, set main conclusions in context of impact to research and society Zeyer et al. (2021) peaky ctc, ambiguous phone boundaries

[20] Extensions from previous research (yang2022) Liu et al. (2023) wav2vec which does adversarial training to improve discrimination. Peng et al. (2023) contrastive loss optimization Lonergan et al. (2024) alternative architectures for low resource asr Neri et al. (2008) pronunciation training for children, lead to improvements comparable to traditional training Prabhavalkar et al. (2017) alternatives to ctc

8 Future work

The absence of a DPAD equivalent in our work, as noted above, suggests a concrete extension. Diversified output for an ASR approach analogous to our own might be achieved by means of shallow fusion of a Language Model (LM) together with Beam Search, whereby a scaling α value weighing the LM contribution might distort English ASR transcriptions of Irish speech to better adhere to English phonotactics, mimicking the sort of phonetic assimilation humans do when exposed to unfamiliar speech (For an example of L1 interference on L2 perception, see Kartushina and Frauenfelder, 2013). With the LM potentially acting to steer outputs preferentially towards English phonotactics, we could diversify the outputs via a Diverse Beam Search (Vijayakumar et al., 2018) similar to that used in D. Zhang et al. (2022). This expanded ASR approach might be preferable for low-resource languages like Irish where expertly annotated L1-L2 word pairs and L2 knowledge databases detailing common phoneme confusions are unavailable. Whether the combination of Irish speech through an English ASR with an English LM approximates realistic phoneme confusions is an empirical question, but a concrete one worth exploring. Even outside potential application in a data synthesis pipeline, the ability to approximate cross-lingual confusion pairs with only monolingual data could be useful for a number of applications.

[21] Suprasegmental eval As mentioned in the background2, suprasegmental features can play an intrumental role in intelligability and is well worth exploring as an object of evaluation. Much of previous work focuses on segmentals as we do here expand. (Thomson and Derwing, 2014) Previous work has approached aspects of suprasegmental pronunciation like prosody as both a (Gong et al., 2022) (Chen and H. Li, 2016)

[22] placeholder: code switching multilingual asr models https://developer.nvidia.com/blog/multilingual-and-code-switched-automatic-speech-recognition-with-nvidia-nemo/

[23] ensemble across model types Due to the relative size of modern models like Wav2Vec2, there is a practical limit to the effective size of a model ensemble where either memory demands or runtime scale linearly with respect to model numbers. Gitman et al. (2023)double check wording does however find that it is possible to extract useful confidence measures from intermediate layers, reducing the runtime cost of adding new models by 4.5 times in their case, which could seems to a promising strategy to make inference times of larger ensembles more managable. If some members of our ensemble are only used in constrained settings (i.e. to determine palatalization as for our Russian model), it might be advantageous to explore more lightweight alternatives focused on discriminating a more limited set of features following Ananthakrishnan et al. (2011). This could be combining Wav2Vec2 models of different parameter size, if we were to combine the 2 billion parameter version with the 300 million parameter version, or it could be of entirely different architectures, combining, for example, Support Vector Machines to discern some targeted features, while relying on a 2 billion parameter Wav2Vec2 model as the work-horse for easier-to-distinguish features or those of more marginal concern.

[24] calibrating the entropy balancing, training a more granular model selector block, full ensemble training

To maximally avoid the need for expensive L2 training data, we opted for a simple heuristic confidence ensemble without using parametric entropy measures, but performance would likely be improved by implementing more complex variants suggested by Laptev and Ginsburg (2023), specifically *tsallis entropy* which would give some control over the sensitivity of the entropy calculation with an *alpha* parameter. This measure becomes Gibbs entropy when alpha is 1, and acts similarly to temperature scaling when between 0 and 1. Another realistic extension to the current work could be to train a model selection block using the entropy-based confidence calculations explored following Gitman et al. (2023). The authors also explored a variety of entropy measures which could also be worth exploring for our application, though we confined ourselves here to Gibbs to highlight the proof of concept.

[25] forced alignment for CTC Though we use the CTC forced alignment method in torch audio, this is marked as deprecated for 2.8 and beyond, though there exists alternative alignment methods. Due to the ambiguous calculation of spans by our method used elaborate on this, it would be worth exploring alternatives that may work better for CTC trained models, even perhaps turning to weighted finite-state transducer (WFST)-based alternatives like MFA to perform forced alignment, calculating the Wav2Vec2 frame spans from MFA outputs.

[26] iterative improvement of neural translation model Though in the current work we pursue a one-shot paraphrasing of Irish L1 speech to Irish L2 approximations via a monolingual English ASR model, it could be worth using this initial paraphrasing as a seed corpus to a Seq2Seq paraphraser analogous to D. Zhang et al. (2022). At the cost of some expert annotation, this initial corpus could be further used in a supervised bootstraping of a paraphraser to yield more precise mappings, and if so, the cost-benefit tradeoffs to this approach could be explored to illuminate the space for practical applications.

[27] more advanced contrastive training, tuning word embeddings to target particularly relevant features As segmental MDD is concerned with contrasts between well-pronounced and mispronounced phones common in L2 speech, it might be worth pursuing more advanced contrastive training strategies beyond the adversarial sampling strategy applied here. One interesting avenue might be to investigate possible gains in discriminating high-value contrasts such as palatalization/velarization in Irish.

[28] final eval considerations Finally, lack of readily-available evaluation data left us in the unenviable position of having no reliable data to evaluate the current experiments with. As MDD is a low-resource domain for most languages, low-cost strategies for reliably evaluating such systems for lower-resource languages could make them much more accessible where they are needed most. More systematic investigation of synthetic speech deviations from authentic speech could illuminate how interchangable it is with authentic speech for training of course, but could also illuminate along which dimensions it is a reliable speech target to compare to. Similarly Crosslinguistic comparison of targeted features such as palatalization

start expanding on this to set an upper bound on my experiment instead of letting it scope creep, then make a time machine and give it to yourself 6 months ago Gong et al. (2022) assessment targeting more than one aspect of speech (prosody, word-level stress)

Rouditchenko et al. (2023) language family in pretraining predictive of how models compare. need for resources for smaller families

9 Ethical Considerations

[29] General Ethical considerations for current work and possible extensions
[30] Ethical considerations for Minority languages

10 AI Tools

[31] Describe use of AI tools and how its use benefited me.

Bibliography

Agrawal, Aakriti, Milind Rao, Anit Kumar Sahu, Gopinath Chennupati, and Andreas Stolcke (2023). "Learning When to Trust Which Teacher for Weakly Supervised ASR". In: *INTERSPEECH 2023*. ISCA, Aug. 2023, pp. 381–385. DOI: 10.21437/Interspeech.2023-2205. (Visited on 2025-06-16).

Ananthakrishnan, Gopal, Preben Wik, Olov Engwall, and Sherif Abdou (2011). "Using an Ensemble of Classifiers for Mispronunciation Feedback". In: *Speech and Language Technology in Education (SLaTE 2011)*. ISCA, Aug. 2011, pp. 49–52. DOI: 10.21437/SLaTE.2011-13. (Visited on 2025-09-17).

and others Biewald, Lukas (2020). "Experiment Tracking with Weights and Biases".

Arunkumar, A., Vrunda N. Sukhadia, and S. Umesh (2022). "Investigation of Ensemble Features of Self-Supervised Pretrained Models for Automatic Speech Recognition". In: *Interspeech 2022*. Sept. 2022, pp. 5145–5149. DOI: 10.21437/Interspeech.2022-11376. arXiv: 2206.05518 [cs]. (Visited on 2025-08-29).

Babu, Arun, Changhan Wang, Andros Tjandra, Kushal Lakhotia, Qiantong Xu, Naman Goyal, Kritika Singh, Patrick von Platen, Yatharth Saraf, Juan Pino, Alexei Baevski, Alexis Conneau, and Michael Auli (2021). *XLS-R: Self-supervised Cross-lingual Speech Representation Learning at Scale*. Dec. 2021. DOI: 10.48550/arXiv.2111.09296. arXiv: 2111.09296 [cs]. (Visited on 2025-09-07).

Baevski, Alexei, Michael Auli, and Abdelrahman Mohamed (2020). *Effectiveness of Self-Supervised Pre-Training for Speech Recognition*. May 2020. DOI: 10.48550/arXiv.1911.03912. arXiv: 1911.03912 [cs]. (Visited on 2025-04-04).

Baevski, Alexei, Yuhao Zhou, Abdelrahman Mohamed, and Michael Auli (2020). "Wav2vec 2.0: A Framework for Self-Supervised Learning of Speech Representations". In: *Advances in Neural Information Processing Systems*. Vol. 33. Curran Associates, Inc., pp. 12449–12460. (Visited on 2025-09-09).

Bajorek, Joan Palmiter (2017). "L2 Pronunciation in CALL: The Unrealized Potential of Rosetta Stone, Duolingo, Babbel, and Mango Languages". *Issues and Trends in Educational Technology* 5.1, pp. 24–51.

Bartelds, Martijn, Nay San, Bradley McDonnell, Dan Jurafsky, and Martijn Wieling (2023). *Making More of Little Data: Improving Low-Resource Automatic Speech Recognition Using Data Augmentation*. May 2023. DOI: 10.48550/arXiv.2305.10951. arXiv: 2305.10951 [cs]. (Visited on 2025-07-15).

Bender, Emily M. (2011). "On Achieving and Evaluating Language-Independence in NLP". *Linguistic Issues in Language Technology* 6 (Oct. 2011). ISSN: 1945-3604. DOI: 10.33011/lilt.v6i.1239. (Visited on 2025-04-04).

Broin, Brian Ó (2014). "New Urban Irish: Pidgin, Creole, or Bona Fide Dialect? The Phonetics and Morphology of City and Gaeltacht Speakers Systematically Compared".

Calik, Sukru Selim, Ayhan Kucukmanisa, and Zeynep Hilal Kilimci (2023). *An Ensemble-Based Framework for Mispronunciation Detection of Arabic Phonemes*. Jan. 2023. DOI: 10.48550/arXiv.2301.01378. arXiv: 2301.01378 [cs]. (Visited on 2025-09-06).

Chapelle, Carol A. (2008). "Computer Assisted Language Learning". In: *The Handbook of Educational Linguistics*. Ed. by Bernard Spolsky and Francis M Hult. Blackwell Publishing.

Chasaide, Ailbhe, Neasa Ní Chiaráin, Christoph Wendler, Harald Berthelsen, Andy Murphy, and Christer Gobl (2017). *The ABAIR Initiative: Bringing Spoken Irish into the Digital Space*. Aug. 2017. DOI: 10.21437/Interspeech.2017-1407.

Chen, Nancy F. and Haizhou Li (2016). "Computer-Assisted Pronunciation Training: From Pronunciation Scoring towards Spoken Language Learning". In: *2016 Asia-Pacific Signal and Information Processing Association Annual Summit and Conference (APSIPA)*. Jeju, South Korea: IEEE, Dec. 2016, pp. 1–7. ISBN: 978-988-14768-2-1. DOI: 10.1109/APSIPA.2016.7820782. (Visited on 2025-04-04).

Conneau, Alexis, Alexei Baevski, Ronan Collobert, Abdelrahman Mohamed, and Michael Auli (2020). *Unsupervised Cross-lingual Representation Learning for Speech Recognition*. Dec. 2020. DOI: 10.48550/arXiv.2006.13979. arXiv: 2006.13979 [cs]. (Visited on 2025-04-04).

Conneau, Alexis, Min Ma, Simran Khanuja, Yu Zhang, Vera Axelrod, Siddharth Dalmia, Jason Riesa, Clara Rivera, and Ankur Bapna (2022). *FLEURS: Few-shot Learning Evaluation of Universal Representations of Speech*. May 2022. DOI: 10.48550/arXiv.2205.12446. arXiv: 2205.12446 [cs]. (Visited on 2025-04-04).

Deichler, Anna, Jim O'Regan, and Jonas Beskow (2024). *MM-Conv: A Multi-modal Conversational Dataset for Virtual Humans*. Sept. 2024. DOI: 10.48550/arXiv.2410.00253. arXiv: 2410.00253 [cs]. (Visited on 2025-04-04).

Deng, Li and John C. Platt (2014). "Ensemble Deep Learning for Speech Recognition". In: *Proc. Interspeech*. (Visited on 2025-08-29).

Deng, li, Dong Yu, and John Platt (2012). "Scalable Stacking and Learning for Building Deep Architectures". *Acoustics, Speech, and Signal Processing, 1988. ICASSP-88., 1988 International Conference on* (Mar. 2012). DOI: 10.1109/ICASSP.2012.6288333.

Devlin, Jacob, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova (2019). "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding". In: *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*. Ed. by Jill Burstein, Christy Doran, and Thamar Solorio. Minneapolis, Minnesota: Association for Computational Linguistics, June 2019, pp. 4171–4186. DOI: 10.18653/v1/N19-1423. (Visited on 2025-08-28).

Dietterich, Thomas G (1998). "Approximate Statistical Tests for Comparing Supervised Classification Learning Algorithms". *Neural computation* 10, pp. 1895–1923.

Dietterich, Thomas G (2000). "Ensemble Methods in Machine Learning". In: *International Workshop on Multiple Classifier Systems*. Springer, pp. 1–15. (Visited on 2025-09-06).

Dryer, Matthew, Martin Haspelmath, Matthew Dryer, and Martin Haspelmath (2024). *The World Atlas of Language Structures Online*. Oct. 2024. DOI: 10.5281/ZENODO.13950591. (Visited on 2025-07-08).

Engstrand, Olle (2004). *Fonetikens Grunder*. Lund: Studentlitteratur. ISBN: 978-91-44-04238-1.

Fazel, Amin, Wei Yang, Yulan Liu, Roberto Barra-Chicote, Yixiong Meng, Roland Maas, and Jasha Droppo (2021). *SynthASR: Unlocking Synthetic Data for Speech Recognition*. June 2021. DOI: 10.48550/arXiv.2106.07803. arXiv: 2106.07803 [cs]. (Visited on 2025-09-13).

FFmpeg Team (2024). *FFmpeg*. FFmpeg.

Fiscus, Jonathan G (1997). "A Post-Processing System to Yield Reduced Word Error Rates: Recognizer Output Voting Error Reduction (ROVER)". In: *1997 IEEE Workshop on Automatic Speech Recognition and Understanding Proceedings*. IEEE, pp. 347–354.

Gabriele, Jennifer C (2018). "English Influence on L2 Speakers' Production of Palatalization and Velarization".

Garofolo, John S, Lori F Lamel, William M Fisher, Jonathan G Fiscus, David S Pallett, and Nancy L Dahlgren (1993). *DARPA TIMIT: Acoustic-Phonetic Continuous Speech Corpus CD-ROM, NIST Speech Disc 1-1.1*. Tech. rep. NIST IR 4930. Gaithersburg, MD: National Institute of Standards and Technology, NIST IR 4930. DOI: 10.6028/NIST.IR.4930. (Visited on 2025-09-09).

Gitman, Igor, Vitaly Lavrukhin, Aleksandr Laptev, and Boris Ginsburg (2023). "Confidence-Based Ensembles of End-to-End Speech Recognition Models". In: *INTERSPEECH 2023*. Aug. 2023, pp. 1414–1418. DOI: 10.21437/Interspeech.2023-1281. arXiv: 2306.15824 [eess]. (Visited on 2025-04-04).

Gong, Yuan, Ziyi Chen, Iek-Heng Chu, Peng Chang, and James Glass (2022). "Transformer-Based Multi-Aspect Multi-Granularity Non-Native English Speaker Pronunciation Assessment". In: *ICASSP 2022 - 2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. May 2022, pp. 7262–7266. DOI: 10.1109/ICASSP43922.2022.9746743. arXiv: 2205.03432 [cs]. (Visited on 2025-04-04).

Google Cloud (2026). *Text-to-Speech AI*.

Graves, Alex, Santiago Fernandez, Faustino Gomez, and Jurgen Schmidhuber (n.d.). "Connectionist Temporal Classification: Labelling Unsegmented Sequence Data with Recurrent Neural Networks" ().

Hansen, L.K. and P. Salamon (1990). "Neural Network Ensembles". *IEEE Transactions on Pattern Analysis and Machine Intelligence* 12.10 (Oct. 1990), pp. 993–1001. ISSN: 01628828. DOI: 10.1109/34.58871. (Visited on 2025-09-14).

Hardison, Debra M. (2005). "Second-Language Spoken Word Identification: Effects of Perceptual Training, Visual Cues, and Phonetic Environment". *Applied Psycholinguistics* 26.4 (Oct. 2005), pp. 579–596. ISSN: 0142-7164, 1469-1817. DOI: 10.1017/S0142716405050319. (Visited on 2025-04-04).

Hayashi, Tomoki, Shinji Watanabe, Yu Zhang, Tomoki Toda, Takaaki Hori, Ramon Astudillo, and Kazuya Takeda (2018). *Back-Translation-Style Data Augmentation for End-to-End ASR*. July 2018. DOI: 10.48550/arXiv.1807.10893. arXiv: 1807.10893 [cs]. (Visited on 2025-09-24).

Hedderich, Michael A., Lukas Lange, Heike Adel, Jannik Strötgen, and Dietrich Klakow (2021). *A Survey on Recent Approaches for Natural Language Processing in Low-Resource Scenarios*. Apr. 2021. DOI: 10.48550/arXiv.2010.12309. arXiv: 2010.12309 [cs]. (Visited on 2025-04-04).

Joshi, Pratik, Sebastin Santy, Amar Budhiraja, Kalika Bali, and Monojit Choudhury (2021). *The State and Fate of Linguistic Diversity and Inclusion in the NLP World*. Jan. 2021. DOI: 10.48550/arXiv.2004.09095. arXiv: 2004.09095 [cs]. (Visited on 2025-04-04).

Jurafsky, Dan and James H. Martin (2025). *Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition, with Language Models*. 3rd.

Kartushina, Natalia and Ulrich Hans Frauenfelder (2013). "On the Role of L1 Speech Production in L2 Perception: Evidence from Spanish Learners of French". In: *Interspeech 2013*. ISCA, Aug. 2013, pp. 2118–2122. DOI: 10.21437/Interspeech.2013-502. (Visited on 2026-05-23).

Kheir, Yassine El, Ahmed Ali, and Shammur Absar Chowdhury (2023). *Automatic Pronunciation Assessment – A Review*. Oct. 2023. DOI: 10.48550/arXiv.2310.13974. arXiv: 2310.13974 [cs]. (Visited on 2025-04-04).

Khosla, Prannay, Piotr Teterwak, Chen Wang, Aaron Sarna, Yonglong Tian, Phillip Isola, Aaron Maschinot, Ce Liu, and Dilip Krishnan (2021). *Supervised Contrastive Learning*. Mar. 2021. DOI: 10.48550/arXiv.2004.11362. arXiv: 2004.11362 [cs]. (Visited on 2025-10-02).

Korzekwa, Daniel, Jaime Lorenzo-Trueba, Thomas Drugman, Shira Calamaro, and Bozena Kostek (2021). *Weakly-Supervised Word-Level Pronunciation Error Detection in Non-Native English Speech*. June 2021. DOI: 10.48550/arXiv.2106.03494. arXiv: 2106.03494 [eess]. (Visited on 2025-09-25).

Korzekwa, Daniel, Jaime Lorenzo-Trueba, Thomas Drugman, and Bozena Kostek (2022). "Computer-Assisted Pronunciation Training—Speech Synthesis Is Almost All You Need". *Speech Communication* 142 (July 2022), pp. 22–33. ISSN: 0167-6393. DOI: 10.1016/j.specom.2022.06.003. (Visited on 2025-07-18).

Krashen, Stephen D. (1984). *Principles and Practice in Second Language Acquisition*. Reprinted. Language Teaching Methodology Series. Oxford: Pergamon Press. ISBN: 978-0-08-028628-0.

Laptev, Aleksandr and Boris Ginsburg (2023). "Fast Entropy-Based Methods of Word-Level Confidence Estimation for End-To-End Automatic Speech Recognition". In: *2022 IEEE Spoken Language Technology Workshop (SLT)*. Jan. 2023, pp. 152–159. DOI: 10.1109/SLT54892.2023.10022960. arXiv: 2212.08703 [eess]. (Visited on 2025-09-14).

Leung, Wai-Kim, Xunying Liu, and Helen Meng (2019). "CNN-RNN-CTC Based End-to-end Mispronunciation Detection and Diagnosis". In: *ICASSP 2019 - 2019 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. Brighton, United Kingdom: IEEE, May 2019, pp. 8132–8136. ISBN: 978-1-4799-8131-1. DOI: 10.1109/ICASSP.2019.8682654. (Visited on 2025-09-12).

Li, Kun, Xiaojun Qian, and Helen Meng (2017). "Mispronunciation Detection and Diagnosis in L2 English Speech Using Multidistribution Deep Neural Networks". *IEEE/ACM Transactions on Audio, Speech, and Language Processing* 25.1 (Jan. 2017), pp. 193–207. ISSN: 2329-9290, 2329-9304. DOI: 10.1109/TASLP.2016.2621675. (Visited on 2025-04-04).

Liu, Alexander H., Wei-Ning Hsu, Michael Auli, and Alexei Baevski (2023). "Towards End-to-End Unsupervised Speech Recognition". In: *2022 IEEE Spoken Language Technology Workshop (SLT)*. Doha, Qatar: IEEE, Jan. 2023, pp. 221–228. ISBN: 979-8-3503-9690-4. DOI: 10.1109/SLT54892.2023.10023187. (Visited on 2025-04-04).

Lo, Tien-Hong, Shi-Yan Weng, Hsiu-Jui Chang, and Berlin Chen (2020). *An Effective End-to-End Modeling Approach for Mispronunciation Detection*. May 2020. DOI: 10.48550/arXiv.2005.08440. arXiv: 2005.08440 [eess]. (Visited on 2025-08-30).

Lonergan, Liam, Mengjie Qian, Neasa Ní Chiaráin, Christer Gobl, and Ailbhe Ní Chasaide (2024). *Low-Resource Speech Recognition and Dialect Identification of Irish in a Multi-Task Framework*. May 2024. DOI: 10.48550/arXiv.2405.01293. arXiv: 2405.01293 [cs]. (Visited on 2025-04-04).

Maclin, Richard and David Opitz (1997). "An Empirical Evaluation of Bagging and Boosting". *AAAI/IAAI* 1997, pp. 546–551. (Visited on 2025-09-17).

Magueresse, Alexandre, Vincent Carles, and Evan Heetderks (2020). *Low-Resource Languages: A Review of Past Work and Future Challenges*. June 2020. DOI: 10.48550/arXiv.2006.07264. arXiv: 2006.07264 [cs]. (Visited on 2025-04-04).

Mortensen, David R, Patrick Littell, Akash Bharadwaj, Kartik Goyal, Chris Dyer, and Lori Levin (n.d.). "PanPhon: A Resource for Mapping IPA Segments to Articulatory Feature Vectors" ().

Moseley, Christopher and Alexandre Nicolas (2010). *Atlas of the World's Languages in Danger*. Paris.

Needleman, Saul B and Christian D Wunsch (1970). "A General Method Applicable to the Search for Similarities in the Amino Acid Sequence of Two Proteins". *Journal of molecular biology* 48.3, pp. 443–453.

Neri, Ambra, Ornella Mich, Matteo Gerosa, and Diego Giuliani (2008). "The Effectiveness of Computer Assisted Pronunciation Training for Foreign Language Learning by Children". *Computer Assisted Language Learning* 21.5 (Dec. 2008), pp. 393–408.

ISSN: 0958-8221, 1744-3210. DOI: 10.1080/09588220802447651. (Visited on 2025-04-04).

Ní Chasaide, Ailbhe, Neasa Ní Chiaráin, Harald Berthelsen, Christoph Wendler, and Andrew Murphy (2015). "SPEECH TECHNOLOGY AS DOCUMENTATION FOR ENDANGERED LANGUAGE PRESERVATION: THE CASE OF IRISH".

Ní Chasaide, Ailbhe, Neasa Ní Chiaráin, Harald Berthelsen, Christoph Wendler, Andrew Murphy, Emily Barnes, and Christer Gobl (2019). "Can We Defuse the Digital Timebomb? Linguistics, Speech Technology and the Irish Language Community". In: *Proceedings of the Language Technologies for All (LT4All)*. European Language Resources Association (ELRA), pp. 177–181. DOI: 10.21437/SpeechProsody.2016-73. (Visited on 2025-04-04).

Ní Chiaráin, Neasa (2022). "An Corpas Cliste: Creating a Learner Corpus for Irish from a New, Purpose-Built iCALL Platform". In: *Intelligent CALL, Granular Systems and Learner Data: Short Papers from EUROCALL 2022*. Ed. by Birna Arnbjörnsdóttir, Branislav Bédi, Linda Bradley, Kolbrún Friðriksdóttir, Hólmfríður Garðarsdóttir, Sylvie Thouësny, and Matthew James Whelpton. 1st ed. Research-publishing.net, Dec. 2022, pp. 297–301. ISBN: 978-2-38372-015-7. DOI: 10.14705/rpnet.2022.61.1474. (Visited on 2025-04-04).

Ní Chiaráin, Neasa, Oisín Nolan, Madeleine Comtois, Neimhin Robinson Gunning, Harald Berthelsen, and Ailbhe Ni Chasaide (2022). "Using Speech and NLP Resources to Build an iCALL Platform for a Minority Language, the Story of An Scéalaí, the Irish Experience to Date". In: *Proceedings of the Fifth Workshop on the Use of Computational Methods in the Study of Endangered Languages*. Dublin, Ireland: Association for Computational Linguistics, pp. 109–118. DOI: 10.18653/v1/2022.computel-1.14. (Visited on 2025-04-04).

Parikh, Aditya Kamlesh, Cristian Tejedor-Garcia, Catia Cucchiarini, and Helmer Strik (2025). *Evaluating Logit-Based GOP Scores for Mispronunciation Detection*. July 2025. DOI: 10.48550/arXiv.2506.12067. arXiv: 2506.12067 [eess]. (Visited on 2025-08-31).

Paszke, Adam, Sam Gross, Francisco Massa, Adam Lerer, James Bradbury, Gregory Chanan, Trevor Killeen, Zeming Lin, Natalia Gimelshein, Luca Antiga, Alban Desmaison, Andreas Kopf, Edward Yang, Zachary DeVito, Martin Raison, Alykhan Tejani, Sasank Chilamkurthy, Benoit Steiner, Lu Fang, Junjie Bai, and Soumith Chintala (2019). "PyTorch: An Imperative Style, High-Performance Deep Learning Library". In: *Advances in Neural Information Processing Systems*. Vol. 32. Curran Associates, Inc. (Visited on 2025-09-08).

Peng, Linkai, Yingming Gao, Rian Bao, Ya Li, and Jinsong Zhang (2023). "End-to-End Mispronunciation Detection and Diagnosis Using Transfer Learning". *Applied Sciences* 13.11 (June 2023), p. 6793. ISSN: 2076-3417. DOI: 10.3390/app13116793. (Visited on 2025-06-16).

Povey, Daniel, Arnab Ghoshal, Gilles Boulianne, Lukaˇs Burget, Ondˇrej Glembek, Nagendra Goel, Mirko Hannemann, Petr Motlıˇcek, Yanmin Qian, Petr Schwarz, Jan Silovsky, Georg Stemmer, and Karel Vesely (2011). "The Kaldi Speech Recognition Toolkit".

Prabhavalkar, Rohit, Tara N. Sainath, Yonghui Wu, Patrick Nguyen, Zhifeng Chen, Chung-Cheng Chiu, and Anjuli Kannan (2017). *Minimum Word Error Rate Training for Attention-based Sequence-to-Sequence Models*. Dec. 2017. DOI: 10.48550/arXiv.1712.01818. arXiv: 1712.01818 [cs]. (Visited on 2025-04-04).

Qian, Xiaojun, Frank K. Soong, and Helen Meng (2010). "Discriminative Acoustic Model for Improving Mispronunciation Detection and Diagnosis in Computer-Aided Pronunciation Training (CAPT)". In: *Interspeech 2010*. ISCA, Sept. 2010, pp. 757–760. DOI: 10.21437/Interspeech.2010-278. (Visited on 2026-05-15).

Ranathunga, Surangika, En-Shiun Annie Lee, Marjana Prifti Skenduli, Ravi Shekhar, Mehreen Alam, and Rishemjit Kaur (2021). *Neural Machine Translation for Low-Resource Languages: A Survey*. June 2021. DOI: 10.48550/arXiv.2106.15115. arXiv: 2106.15115 [cs]. (Visited on 2025-04-04).

Raschka, Sebastian (2018). "MLxtend: Providing Machine Learning and Data Science Utilities and Extensions to Python's Scientific Computing Stack". *Journal of Open Source Software* 3.24 (Apr. 2018), p. 638. ISSN: 2475-9066. DOI: 10.21105/joss.00638. (Visited on 2026-05-19).

Rice, Lloyd (1976). "Hardware and Software for Speech Synthesis". 1, pp. 6–8. (Visited on 2026-05-17).

Rogerson-Revell, Pamela M (2021). "Computer-Assisted Pronunciation Training (CAPT): Current Issues and Future Directions". *RELC Journal* 52.1 (Apr. 2021), pp. 189–205. ISSN: 0033-6882, 1745-526X. DOI: 10.1177/0033688220977406. (Visited on 2025-04-04).

Rosenblum, Lawrence D. (2008). "Speech Perception as a Multimodal Phenomenon". *Current Directions in Psychological Science* 17.6 (Dec. 2008), pp. 405–409. ISSN: 0963-7214, 1467-8721. DOI: 10.1111/j.1467-8721.2008.00615.x. (Visited on 2025-04-04).

Rouditchenko, Andrew, Sameer Khurana, Samuel Thomas, Rogerio Feris, Leonid Karlinsky, Hilde Kuehne, David Harwath, Brian Kingsbury, and James Glass (2023). "Comparison of Multilingual Self-Supervised and Weakly-Supervised Speech Pre-Training for Adaptation to Unseen Languages". In: *INTERSPEECH 2023*. ISCA, Aug. 2023, pp. 2268–2272. DOI: 10.21437/Interspeech.2023-1061. (Visited on 2025-04-04).

Snesareva, Marina (2016). "Palatalization in Dublin Irish: The Extent of Phonetic Interference". *Procedia - Social and Behavioral Sciences* 236 (Dec. 2016), pp. 213–218. ISSN: 18770428. DOI: 10.1016/j.sbspro.2016.12.009. (Visited on 2025-04-04).

Someki, Masao, Kwanghee Choi, Siddhant Arora, William Chen, Samuele Cornell, Jionghao Han, Yifan Peng, Jiatong Shi, Vaibhav Srivastav, and Shinji Watanabe (2024). *ESPnet-EZ: Python-only ESPnet for Easy Fine-tuning and Integration*. Sept. 2024. DOI: 10.48550/arXiv.2409.09506. arXiv: 2409.09506 [cs]. (Visited on 2025-04-04).

Stenson, Nancy (2020). *Modern Irish: A Comprehensive Grammar*. Routledge Comprehensive Grammars. London ; New York: Routledge, Taylor & Francis. ISBN: 978-1-138-23652-3 978-1-138-23651-6.

Thai, Bao, Robert Jimerson, Dominic Arcoraci, Emily Prud'hommeaux, and Raymond Ptucha (2019). "Synthetic Data Augmentation for Improving Low-Resource ASR". In: *2019 IEEE Western New York Image and Signal Processing Workshop (WNYISPW)*. Rochester, NY, USA: IEEE, Oct. 2019, pp. 1–9. ISBN: 978-1-7281-4352-1. DOI: 10.1109/WNYIPW.2019.8923082. (Visited on 2025-08-28).

Thomson, Ron and Tracey Derwing (2014). "The Effectiveness of L2 Pronunciation Instruction: A Narrative Review". *Applied Linguistics* 2014 (Dec. 2014), pp. 1–20. DOI: 10.1093/applin/amu076.

van den Oord, Aaron, Sander Dieleman, Heiga Zen, Karen Simonyan, Oriol Vinyals, Alex Graves, Nal Kalchbrenner, Andrew Senior, and Koray Kavukcuoglu (2016). *WaveNet: A Generative Model for Raw Audio*. Sept. 2016. DOI: 10.48550/arXiv.1609.03499. arXiv: 1609.03499 [cs.SD]. (Visited on 2026-05-17).

Vijayakumar, Ashwin K., Michael Cogswell, Ramprasath R. Selvaraju, Qing Sun, Stefan Lee, David Crandall, and Dhruv Batra (2018). *Diverse Beam Search: Decoding Diverse Solutions from Neural Sequence Models*. Oct. 2018. DOI: 10.48550/arXiv.1610.02424. arXiv: 1610.02424 [cs.AI]. (Visited on 2026-05-23).

Watanabe, Shinji, Takaaki Hori, Shigeki Karita, Tomoki Hayashi, Jiro Nishitoba, Yuya Unno, Nelson Enrique Yalta Soplin, Jahn Heymann, Matthew Wiesner, Nanxin Chen, Adithya Renduchintala, and Tsubasa Ochiai (2018). *ESPnet: End-to-End Speech Pro-*

cessing Toolkit. Mar. 2018. DOI: 10.48550/arXiv.1804.00015. arXiv: 1804.00015 [cs]. (Visited on 2025-09-03).
Wei, Hongxin, Renchunzi Xie, Hao Cheng, Lei Feng, Bo An, and Yixuan Li (2022). *Mitigating Neural Network Overconfidence with Logit Normalization*. June 2022. DOI: 10.48550/arXiv.2205.09310. arXiv: 2205.09310 [cs]. (Visited on 2025-06-16).
Witt, Silke and Steve Young (2014). "Computer-Assisted Pronunciation Teaching Based on Automatic Speech Recognition". In: *Language Teaching and Language Technology*. Routledge, pp. 25–35.
Witt, Silke M (n.d.). "Automatic Error Detection in Pronunciation Training: Where We Are and Where We Need to Go" ().
Witt, Silke Maren (2000). "Use of Speech Recognition in Computer-Assisted Language Learning." PhD thesis.
Wolf, Thomas, Lysandre Debut, Victor Sanh, Julien Chaumond, Clement Delangue, Anthony Moi, Pierric Cistac, Tim Rault, Remi Louf, Morgan Funtowicz, Joe Davison, Sam Shleifer, Patrick von Platen, Clara Ma, Yacine Jernite, Julien Plu, Canwen Xu, Teven Le Scao, Sylvain Gugger, Mariama Drame, Quentin Lhoest, and Alexander Rush (2020). "Transformers: State-of-the-Art Natural Language Processing". In: *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*. Ed. by Qun Liu and David Schlangen. Online: Association for Computational Linguistics, Oct. 2020, pp. 38–45. DOI: 10.18653/v1/2020.emnlp-demos.6. (Visited on 2025-09-08).
Wu, Minglin, Kun Li, Wai-Kim Leung, and Helen Meng (2021). "Transformer Based End-to-End Mispronunciation Detection and Diagnosis". In: *Interspeech 2021*. ISCA, Aug. 2021, pp. 3954–3958. DOI: 10.21437/Interspeech.2021-1467. (Visited on 2025-04-04).
Xu, Qiantong, Tatiana Likhomanenko, Jacob Kahn, Awni Hannun, Gabriel Synnaeve, and Ronan Collobert (2020). "Iterative Pseudo-Labeling for Speech Recognition". In: *Interspeech 2020*. ISCA, Oct. 2020, pp. 1006–1010. DOI: 10.21437/Interspeech.2020-1800. (Visited on 2025-04-04).
Yang, Mu, Kevin Hirschi, Stephen Daniel Looney, Okim Kang, and John H.L. Hansen (2022). "Improving Mispronunciation Detection with Wav2vec2-based Momentum Pseudo-Labeling for Accentedness and Intelligibility Assessment". In: *Interspeech 2022*. ISCA, Sept. 2022, pp. 4481–4485. DOI: 10.21437/Interspeech.2022-11039. (Visited on 2025-04-04).
Yu, Dong and Li Deng (2015). *Automatic Speech Recognition: A Deep Learning Approach*. Signals and Communication Technology. London: Springer London. ISBN: 978-1-4471-5778-6 978-1-4471-5779-3. DOI: 10.1007/978-1-4471-5779-3. (Visited on 2025-08-31).
Zeyer, Albert, Ralf Schlüter, and Hermann Ney (2021). *Why Does CTC Result in Peaky Behavior?* June 2021. DOI: 10.48550/arXiv.2105.14849. arXiv: 2105.14849 [cs]. (Visited on 2025-06-16).
Zhang, Daniel, Ashwinkumar Ganesan, Sarah Campbell, and Daniel Korzekwa (2022). "L2-GEN: A Neural Phoneme Paraphrasing Approach to L2 Speech Synthesis for Mispronunciation Diagnosis". In: *Interspeech 2022*. ISCA, Sept. 2022, pp. 4317–4321. DOI: 10.21437/Interspeech.2022-209. (Visited on 2025-04-04).
Zhang, Junbo, Zhiwen Zhang, Yongqing Wang, Zhiyong Yan, Qiong Song, Yukai Huang, Ke Li, Daniel Povey, and Yujun Wang (2021). *Speechocean762: An Open-Source Non-native English Speech Corpus For Pronunciation Assessment*. June 2021. DOI: 10.48550/arXiv.2104.01378. arXiv: 2104.01378 [cs]. (Visited on 2025-04-04).
Zhao, Guanlong, Sinem Sonsaat, Alif Silpachai, Ivana Lucic, Evgeny Chukharev-Hudilainen, John Levis, and Ricardo Gutierrez-Osuna (2018). "L2-ARCTIC: A Non-native En-

glish Speech Corpus". In: *Interspeech 2018*. ISCA, Sept. 2018, pp. 2783–2787. DOI: 10.21437/Interspeech.2018-1110. (Visited on 2025-04-04).